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A strategic guide of techniques for biomedical and tissue engineering applications to measure mechanical properties of soft matter, eye and skin

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GRAPHICAL ABSTRACT

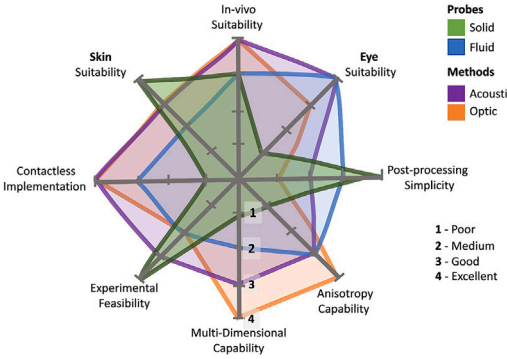


Table-of-Content: Radar plot showing the main capabilities of each type of measurement technique. We grouped the techniques in four big groups: contacting solid methods, contacting liquid methods, acoustic methods and optical methods.

ABSTRACT

A comprehensive understanding of tissue mechanics at the microscale is critical for advancing personalised therapies, controlled drug release, and tissue engineering. Characterising the mechanical properties of complex, soft biological materials—particularly multilayered and anisotropic organs such as the skin and eye—remains a significant challenge due to their variable water content and scale-dependent behaviour. Traditional continuum models and linear material responses often fail to capture the dynamic and nonlinear nature of these tissues under physiologically relevant conditions.

This review provides a strategic overview of state-of-the-art techniques for probing the mechanical properties of soft biological tissues, with a focus on skin and ocular systems. Our focus on the skin and eye reflects their favourable barrier properties for topical drug delivery. We examine visualisation methods including optical imaging, interferometry, digital image correlation, optical coherence microscopy, and acoustic imaging. In parallel, we assess actuation mechanisms such as indentation, cavitation rheology, and flow elastography, highlighting their suitability for *in vivo* applications. Each technique is benchmarked against key operational parameters—spatial resolution, acquisition rate, invasiveness, and strain rate—relevant to drug delivery and therapeutic engineering.

By mapping the landscape of mechanical characterisation tools, this work offers a valuable resource for researchers in biomedical engineering and beyond, including fields such as physics and chemistry, where accurate dynamic analysis of soft complex materials is essential.

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1. Introduction

In the bioengineering and medical fields, a deep understanding of the mechanical properties of materials, such as their elasticity, strength, and deformation characteristics, is essential [1–3]. Materials used for implants, medical devices, and tissue engineering must have specific and reliable mechanical properties to integrate into biological systems [4–6]. Precise knowledge of the material's behaviour guides the design and optimisation of prosthetic devices, ensuring optimal functionality and patient comfort [7]. In addition, controlling material properties is vital in wound healing, where the mechanical properties of dressings and scaffolds directly influence the healing process by affecting factors such as cell migration and tissue remodelling [5,8,9]. For these applications, it is necessary to know a priori, and ideally predict, how the mechanical behaviour of tissues changes due to ageing or external ambient conditions. By extension, understanding how soft tissues respond to rapid, localised loading is also fundamental to the design and safe operation of modern drug delivery modalities, and could lead to personalised treatments and targeted drug delivery in skin and ocular tissues, as well as other soft tissues [5,10,11].

Advanced drug-delivery methods, ranging from microneedles and tattoo-needles to gene guns, (micro)jet injectors, and ultrasound-assisted approaches, are designed to deliver therapeutic agents with higher spatial precision, reduced pain, and tighter control over depth and dose than conventional needles [12,13]. These techniques can be characterised by their actuation mechanism, temporal dynamics (e.g., puncture rate and injection duration), spatial footprint left in the tissue after delivery (e.g., tip/orifice or jet diameter), achievable penetration depth, and typical dose or volume per delivery event [10]. Fig. 1 summarises the main delivery methods by their injection duration and typical penetration depth. From left to right, the figure shows delivery timescales spanning several orders of magnitude—from microsecond-long jet and microjet injections, through millisecond jet and tattoo methods, to applications lasting seconds or minutes with conventional needles and microneedles. We roughly categorise rapid delivery methods as those operating within less than one second, and superficial injections as those operating in the first ~1–2 mm into the tissue. In this schematic, conventional needles are included primarily as a reference for standard single-dose subcutaneous/intradermal injections, but the timescale range also encompasses other needle-based systems such as continuous subcutaneous insulin infusion pumps and skin-patches, which deliver over hours to days. Table 1 provides a detailed comparison of the delivery methods. It groups them into liquid injectors, ballistic particle systems and solid injectors, and lists their actuation principles. It also reports key temporal metrics such as jet speed, particle or gas speed, or injection time and frequency, as well as spatial parameters including needle, jet or particle size, penetration depth in skin, and typical volume ranges.

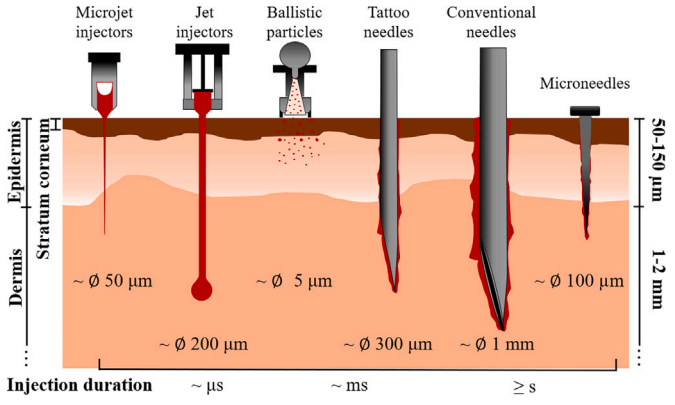


Fig. 1. Representative transdermal delivery techniques compared by injection duration and penetration depth. Approximate thicknesses of the indicated skin layers are: stratum corneum, 10–40 µm; epidermis, 50–150 µm; dermis, 1–2 mm. Rapid liquid-based methods (microjet and jet injectors) and ballistic particles operate on microsecond to millisecond timescales, while solid-needle approaches (tattooing, conventional needles, microneedles) span milliseconds to minutes. These differences illustrate the range of temporal and spatial regimes that motivate the mechanical characterisation window addressed in this review; more details in Table 1.

In parallel to these delivery methods, a broad family of permeation-enhancement strategies has emerged. Classical approaches include chemical enhancers, iontophoresis and electroporation, which increase permeability by altering barrier lipids or creating transient aqueous pathways [31–36]. More recent techniques exploit physical energy, for example sonophoresis and ultrasound or sonic-vibration rollers used in cosmetic and ophthalmic applications [37,38]. Laser and thermal ablation, including fractional laser microporation, can also create controlled microchannels in the stratum corneum [39]. Rather than directly injecting fluid, these methods aim to transiently increase skin permeability and thus provide complementary ways of crossing or modulating the barrier.

1.1. Relevance of tissue composition in determining mechanical properties and transport for drug delivery

A deep understanding of heterogeneous soft matter demands characterising its building blocks across scales—from nanometre-sized extracellular matrix (ECM) components, to cells tens of micrometres wide, to layered tissue structures spanning hundreds of micrometres to millimetres. With these properties defined, models become more predictive and measurements more straightforward to interpret. These tissues exhibit complex rate-dependent, anisotropic and viscoelastic behaviours that influence how drugs diffuse, how cells interact with their microenvironment, and how mechanical cues drive biological responses. For instance, the stiffness of the ECM can regulate cell migration, immune cell infiltration, and therapeutic uptake—critical factors in cancer and regenerative medicine [40].

The ECM is a complex three-dimensional network of proteins and polysaccharides that provides structural and biochemical support to surrounding cells. ECM dynamics influence therapeutic penetration, especially in dense fibrotic or tumour tissues, where it acts as a physical barrier to drug diffusion. ECM components like collagen, elastin, fibronectin, and hyaluronic acid can be targeted to improve drug localisation and retention. Moreover, mechanical disruption (e.g., focused ultrasound), enzymatic degradation, and biomaterial-based carriers are being explored to enhance delivery efficacy [41].

This dynamic remodelling of the ECM is largely driven by fibroblasts and myofibroblasts in the skin [42], and by keratocytes in the cornea [43]. Fibroblasts are the main producers of ECM components such as collagen, fibronectin, and proteoglycans. Modulating fibroblast activity has therefore emerged as a promising strategy to reduce ECM-mediated stiffness, enhance drug penetration, and improve immunotherapy efficacy [44,45]. Direct single-cell techniques that allow quantification of fibroblast stiffness include atomic-force microscopy, micropipette aspiration, magnetic/optical tweezers, magnetic twisting cytometry and traction-force microscopy [46–50]. These techniques operate at micron spatial scales and are ideal to probe cell mechanics and mechanotransduction. By contrast, methods at the millimetre scale and those capable of sub-100 ms deformation interrogate tissue-level responses in which fibroblast activity is only one contributor. For example, high-speed indentation/impact testing, instrumented micro-indentation, shear-wave elastography and optical-coherence elastography have been used in this context [51–55]. Therefore, we treat cell-scale techniques as complementary to the main focus of this review and indicate where cell-scale studies can inform interpretation of tissue-level measurements.

Beyond ECM composition, higher-order structural and physiological factors critically shape mechanics and transport. Fibre orientation, crimp and lamellar architecture produce strong anisotropy and strain-stiffening, altering tissue fracture thresholds and preferred penetration

Table 1
Comparison of selected delivery techniques presented in Fig. 1 and their corresponding temporal dynamics, spatial range, and delivered volume range.

Delivery method	Actuation rate	Spatial range (size/depth)	Volume range	Ref.
Liquid injectors				
Jet injectors:				
Spring-driven	60–140 m/s	76–500 μm/2–10 mm	100–1000 μl	[10,14,15]
Gas-pressurised	100–300 m/s	100–300 μm/2–10 mm	>50 μl	[10,16]
Voice coil actuated	50–200 m/s	150–300 μm/1–20 mm	20–1000 μl	[10,17]
Piezoelectric	50–160 m/s	40–130 μm/1–10 mm	10–100 nl	[10,18]
Microjet injectors:				
Laser pulsed	10–850 m/s	100–1400 μm/0.1–5 mm	10–600 nl	[10,19]
Continuous-wave	1–94 m/s	35–120 μm/0.1–0.4 mm	10–1000 nl	[10,20]
Ballistic particles				
Classical GeneGun	430 m/s (gas) 35 m/s (particles)	4 μm particles/15 μm	2 μl (DNA)	[24]
GeneGun (3D barrel)	380 m/s (gas) 35 m/s (particles)	600 nm particles/35 μm	6 μl (gold particles)	[25]
Epidermal Powder Immunisation	160–640 m/s	0.53–89 μm particles/<200 μm	5 ml gas/1 mg powder	[26,27]
Solid injectors				
Conventional needles	~s	0.23–1.27 mm/3–16 mm	0.1–1 ml	[11]
Tattoo needles	50–150 Hz	200–350 μm/0.5–2 mm	< 10 nl	[11,28]
Microneedles	minutes	50–300 μm/0.1–1 mm	120–320 μl	[29,30]

directions [56–58]. Layered organisation imposes serial barriers with distinct moduli and failure modes, so the effective response depends on which layers carry load and at what loading rate [59,60]. Interfacial constituents also matter: the lipid matrix of the stratum corneum and the ocular tear film set surface compliance and friction, influencing device–tissue interaction at the microscale [61,62]. Interstitial spaces and pressure gradients govern load distribution and how fluid and solutes are cleared. In skin, the key drivers are interstitial pressure and perfusion [63]; in the cornea, intraocular pressure and endothelial pump activity which maintain stromal hydration are dominant factors [64].

Together, tissue composition, structure and physiological state determine stiffness, anisotropy, viscoelastic and poroelastic time scales, and failure thresholds, which have a strong influence on drug penetration depth, in-tissue distribution, retention and safety across delivery modalities [65].

1.2. Mechanical properties fundamentals

Given the role of tissue composition, architecture, and physiological state in shaping mechanics and transport, we now address the descriptors and models used to quantify and predict the mechanical behaviour of these tissues at the scales of interest.

An accurate quantification of the mechanical properties of the material, such as elasticity, viscosity, stiffness, and toughness, is essential to predict and control their structural integrity and performance under applied forces or loads [6,66,67]. A combination of visualisation techniques (Fig. 2(a)) and stimulation methods (Fig. 2(b)) enables measurement and controlled perturbation of tissues, and their output informs material selection and device design (see Fig. 2(c)). One of the most commonly reported descriptors of mechanical behaviour is Young's modulus (E). Due to its conceptual simplicity and ease of understanding, it is often used as a single-number measure of stiffness for linear elastic solids or for solids that exhibit linear elastic behaviour in a restricted stimulus range and timescales [68] (see Fig. 2(c.1)). However, biological soft tissues such as skin and the eye do not generally behave as ideal linear, homogeneous, isotropic elastic materials. In fact, they show nonlinear, anisotropic, time-dependent, and multiphase responses that depend on loading magnitude, direction, and loading rate [69].

For soft biological tissues, it is therefore more informative to describe mechanical behaviour through continuum constitutive models that capture the relevant physics for the experimental timescales and length scales of the given application. Depending on the experimental protocol and the deformation regime, researchers commonly use one or a combination of finite-strain hyperelastic models, linear and nonlinear viscoelastic models, poroelastic and biphasic or fibre-reinforced, anisotropic models [69–72]. These frameworks account for nonlinearity, finite strains, anisotropy, rate dependence, and fluid-structure interactions [73,74]. We also note that surface stresses and elastocapillary effects can significantly alter local mechanical responses for soft solids [75,76].

In this Review, we define a soft solid as a material that can behave either like a solid or like a non-Newtonian fluid, depending on the type of load applied. Soft solids with viscoelastic properties respond to deformation through a combination of elasticity—storing and releasing energy—and viscosity—dissipating energy through internal friction [75,77]. This dissipation arises from molecular rearrangements and resistance to deformation at the microscopic level. Specifically, internal friction in viscoelastic materials refers to the energy lost as heat due to the movement and reconfiguration of polymer chains, the sliding of fibrous networks, or the interactions between structural components within the material [78]. Measuring the visco-elasticity of a material involves employing specialised techniques to capture the time-dependent response of materials after stimuli. By studying this response, the storage modulus G' and the loss modulus G'' can be determined [79] (see Fig. 2(c.2)). The storage modulus G' quantifies the elastic response of a material. It reflects the material's ability to store and recover energy during deformation, and indicates its stiffness or resistance to being deformed. Mathematically, G' is derived from the in-phase component of the stress (or force) response relative to the applied strain (or displacement).

In contrast, the loss modulus G'' describes the viscous response of a material. It represents the amount of energy dissipated as heat during deformation, characterising the material's damping properties. G'' is obtained from the out-of-phase component of the stress (or force) response with respect to the applied strain (or displacement).

The stress applied to the surface of soft solids can dictate mechanical behaviour, compared to stiff materials [75]. In liquids, surface stresses take the form of surface tension, which acts to minimise the surface energy of the liquids. In solids, surface stresses can be compressive and anisotropic [80]. For example, in thin metal films the surface stress along one crystallographic axis may be compressive while along the orthogonal axis it is tensile, causing the film to bend or warp. Surface stress or elastocapillary behaviour can dominate over bulk elasticity if the length scale of the deformation is $\delta \propto \gamma/E \sim 1$, with γ the surface tension of the material [75]. Gels with Young's modulus on the order of kPa, and soft biological tissue and elastomers with Young's modulus on the order of MPa exhibit elastocapillary behaviour [75]. As a result, the

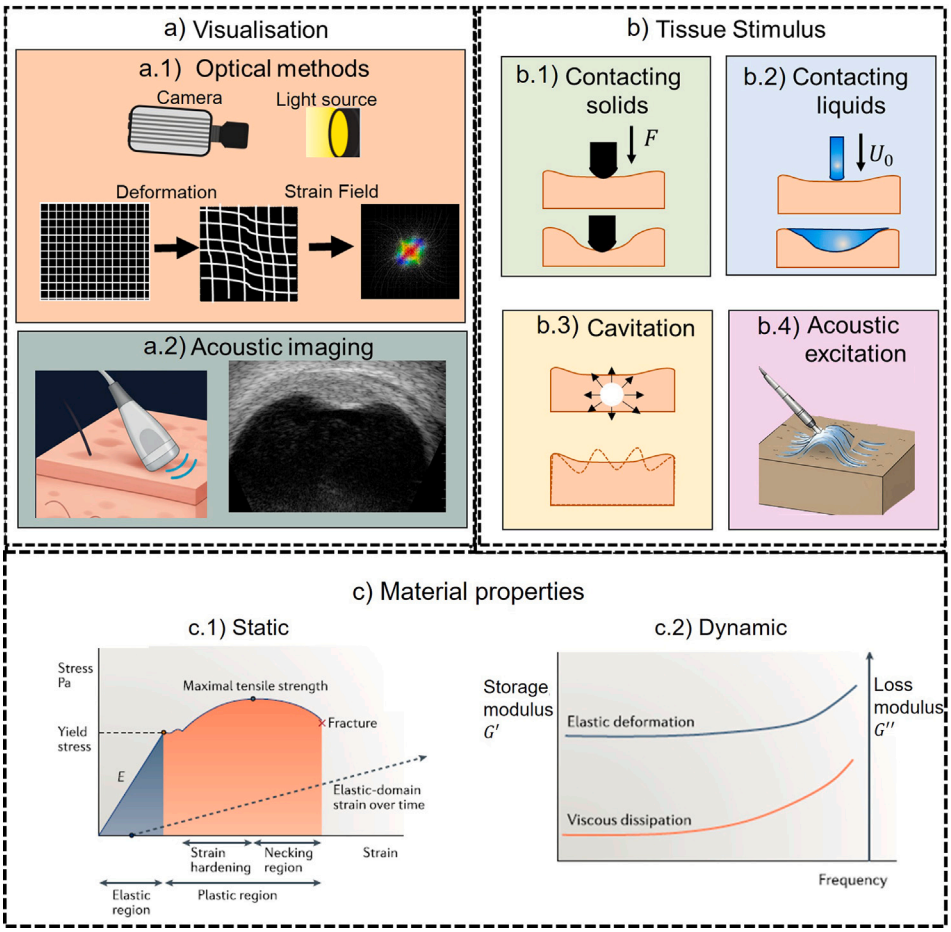


Fig. 2. Methods for Assessing Mechanical Properties of Biological Tissues. This figure summarises the broad categories of methods used to analyse, visualise, and stimulate biological tissues. This review focuses on applying these techniques to understand tissue mechanics at micrometre scales and with rapid deformation, particularly in the eye and skin. (a) Visualisation methods, such as optical and acoustic techniques, that allow for non-invasive observation of tissue deformation. (b) Tissue stimulation methods, employing contacting solid methods, contacting fluid methods, cavitation, and acoustic waves, enable controlled application of forces and stimuli. (c) Material properties and mechanical deformations, including static tensile and shear testing, provide insights into tissue behaviour under stress [Reprinted by permission from reference [6], Copyright, 2020, Springer Nature Limited.].

dominance of elastocapillary effects in soft solids can lead to unexpected deformations or mechanical instabilities, which are critical in the design of biomaterials and flexible devices, especially at small scales.

The need to accurately quantify material properties and predict the behaviour of soft solids has led to the development of novel measurement techniques, ranging from mechanical testing to advanced imaging and spectroscopic techniques [66,81–86]. These methods operate across multiple length scales, and have improved understanding of material property changes from the macroscopic level down to the microstructural and molecular levels. This understanding is crucial because biological tissues have a hierarchical structure and exhibit anisotropic behaviour, meaning their mechanical response varies depending on the length scale and the direction of the applied stress. Consequently, reported mechanical parameters, particularly elastic or shear moduli, span several orders of magnitude ($\approx 10^2$–10^6 Pa), because the measured mechanical output is strongly influenced by the measurement conditions [87,88]. For example, indentation-based skin measurements can differ by up to three orders of magnitude in the values measured with different probe sizes in the same sample [89]. To accurately quantify these properties, techniques must be compared among conflicting requirements, e.g., high sensitivity and high measurement speed. The influence of stimulus variability on the measured mechanical response of samples is not always well understood, posing challenges for standardisation and comparability of results across different laboratories [90–93].

1.3. Bridging material properties and transport in rapid delivery systems

The focus of this review is on measurement approaches directly relevant to modern, fast-acting delivery technologies for skin and ocular tissues. For these technologies to be utilised in practice, an important knowledge gap must be addressed: existing characterisation techniques remain insufficient to correlate intrinsic material properties, such as stiffness, with functional drug permeation outcomes.

Although local stiffness provides information about the mechanical environment experienced during delivery, it is only an indirect predictor of drug permeation in rapid-delivery technologies and must be interpreted in conjunction with device- and tissue-specific dynamics. For instance, in needle-free injection systems, the penetration depth has been shown to depend on both jet velocity and the elastic modulus of the target tissue. Variations in jet velocity can yield markedly different penetration depths even when tissue stiffness remains similar [20]. In practice, accurate prediction of permeation requires consideration of the tissue's dynamic response (viscoelasticity and poroelastic flow) as well as anatomical variability, including thickness, hydration, and pretension [21–23]. Consequently, establishing robust, application-specific correlations demands mechanical mapping at matched strain rates, coupled with calibrated injection trials on appropriate surrogates and *in-situ* verification of deposition (e.g., via labelled tracers or imaging). To date, no existing commercial or reported technology has demonstrated both

the predictive capability and efficiency required to achieve the targeted delivery endpoints.

Whenever possible, we highlight concrete examples where mechanical characterisation has already been used to optimise drug-delivery performance. Simultaneously, we also discuss techniques that have not yet been widely applied in drug-delivery studies but whose spatial and temporal resolution, penetration depth or non-invasive nature make them strong candidates for future use in this context. In this way, the review not only maps out current practice, but also identifies measurement capabilities with potential for guiding the design of next-generation delivery systems, while providing a broader overview that is relevant to the wider community working on soft tissues and biomaterials beyond drug delivery.

Needle-free injection (NFI) and microneedle-based systems operate at scales comparable to cells and tissue microstructures. These modalities aim to improve patient compliance and reduce tissue damage compared to conventional needles, but they impose stringent requirements on how tissue response is measured at the point of delivery. At these scales, the link between mechanics and transport is crucial: mechanical properties govern fluid flow and tissue permeability, influencing penetration depth, lateral spread, and retention time after rapid delivery events [94,95]. Because these properties also modulate active biological processes, predicting delivery outcomes requires measurements that capture the relevant rates and length scales in the actual *in vivo* environment, where perfusion, interstitial pressure, temperature, and cellular activity continuously alter the mechanical state [96].

This is precisely where current measurement techniques fall short. Many widely used methods are optimised for quasi-static loading, bulk samples, or *ex vivo* conditions; few are simultaneously non-invasive, rate-capable, and spatially resolved enough to capture sub-millimetre deformation on sub- to few-millisecond timescales under controlled loading. Closing this technological gap calls for integrating visualisation and actuation strategies that can probe mechanics *in vivo* within the operating window required for advanced delivery technologies. Accordingly, this review consolidates and critically assesses methods that meet these criteria. We focus on experimental approaches operating at spatial scales of 1–500 μm, applying forces of 1–100 mN, and probing material responses on 0.1–100 ms timescales. Each technique is benchmarked by spatial/temporal resolution, invasiveness, and strain-rate capability, and its limitations are identified. Finally, we highlight opportunities where such measurements can inform clinical and research applications—from drug-delivery optimisation to diagnosis—whenever differences in tissue and ECM mechanics are predictive of outcomes.

The review begins with an analysis of the multiscale anatomy of the eye and skin (Section 2). Sections 3–7 systematically evaluate measurement techniques in five categories: optical methods, acoustic methods, contacting solid methods, contacting liquid methods, and cavitation rheology. For each method, we critically compare reported mechanical properties and resolve discrepancies arising from methodological limitations. These insights are synthesised into a decision framework that prioritises selection criteria tailored to dynamic biomedical applications: measurement timescales, spatial resolution, and tissue compatibility. The discussion addresses biological variability such as donor age and sample storage protocols, and emphasises the need for advancing *in situ* measurements to improve precision and ensure effectiveness in drug delivery applications.

2. Basics of the skin and eye anatomy

In this section we describe the basic anatomy components of the eye and skin, which are needed to understand the mechanical properties of these two organs. The cell's structure and inner components lead to the characteristic mechanical properties of both tissues, which we can consider as composite materials. Here, we will focus on the specific parts of the skin and eye that are promising as drug delivery routes, particularly for alternatives to needle-based injections, where a sharp object easily cuts through the tissue.

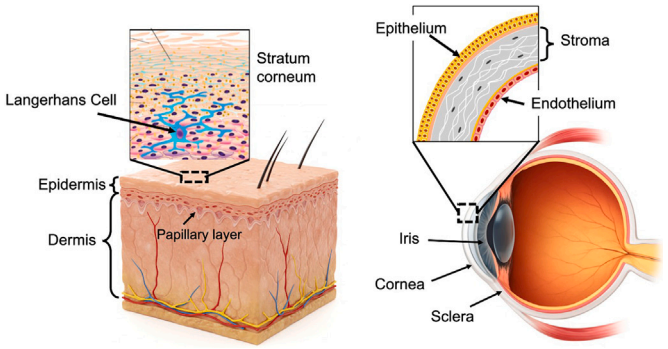


Fig. 3. Left) Schematic cross-sectional representation of human skin (showing the epidermis and the dermis). Right) A simplified schematics of the eye, highlighting the iris, cornea, and sclera. In both cases, we illustrate key anatomical layers relevant to mechanical property measurements discussed in this review.

2.1. Skin main components

The skin has been a preferred site for drug delivery due to its ease of accessibility and large surface area [97,98]. Skin is divided into two layers, the epidermis and dermis (Fig. 3 left). Technically, the hypodermis (2 mm–5 cm) is not considered part of the skin as it mainly consists of connective tissue, however it contributes to its bulk mechanical properties because its constituent adipose tissue has shock-absorbing properties. The dermis (1–2 mm) covers the hypodermis, and is mainly composed of strong connective tissue [99]. The lowest layer (reticular layer) accounts for 80% of the skin thickness, it is an irregular dense network of fibrous connective tissue, complemented with proteoglycans and glycosaminoglycans (GACs), which regulate the visco-elasticity of skin because they can bind water molecules up to 1000 times their own volume. The upper layer of the dermis (papillary layer) maintains the strength and resilience of skin due to its high content of elastin and collagen fibres [100].

The epidermis (50–150 μm) is composed mainly of keratinocytes (≈95%), arranged in a stratified brick and mortar architecture in which keratin and desmosomes act as an intercellular glue, providing mechanical stability and hydrophobicity [101]. Other epidermal cell types (e.g., Merkel cells, melanocytes, Langerhans cells) mainly modulate sensory, pigmentary and immunological functions [12,97,102] and are only indirectly relevant to the mechanical measurements considered in this review.

The epidermis provides the principal skin barrier relevant to both transport and mechanics. The outermost stratum corneum (SC) consists of anucleate corneocytes embedded in a continuous extracellular lipid matrix composed mainly of ceramides, cholesterol and free fatty acids. These lipids are organised into lamellar bilayers whose local composition and packing determine permeability and mechanical properties at the microscale [103]. Beneath the SC, tight junctions in the stratum granulosum form an additional paracellular barrier that contributes to ionic and water retention and to the overall integrity of the epidermal barrier [104,105]. Because the SC lipid matrix is spatially heterogeneous and the junctional architecture is layer-specific, both features strongly influence percutaneous drug penetration and the skin's mechanical response to rapid deformations. These are aspects that need to be considered when selecting techniques that probe millimetre-scale mechanics

and transient (0.1–100 ms) deformations and when interpreting their output [106].

2.2. Eye main components

The eye has been used for drug delivery in e.g., ophthalmology [107–111]. Various drugs, from simple to advanced therapies, are used to treat a wide range of conditions [112,113]. However, designing effective ophthalmic drug delivery systems is challenging, because it must overcome the outer barriers presented by the cornea and the sclera (Fig. 3 right).

The cornea is a highly specialised tissue with thickness $500 \leq \delta \leq 600$ μm [114], and its multilayered structure (particularly the epithelium and stroma) forms a major barrier to ocular drug delivery and mechanical loading [107]. The epithelium is a stratified layer of flat cells, approximately 50 μm thick, that serves as a protective barrier shielding the eye from mechanical injury and preventing pathogens from penetrating into deeper ocular tissues [115–117]. The stroma comprises roughly 90% of the cornea and consists of a dense connective-tissue layer [118]. Unlike the interwoven collagen network of skin, collagen in the corneal stroma is organised into lamellae—sheets of parallel fibrils arranged at varying angles to one another—forming a layered, multidirectional architecture that imparts the tissue its mechanical strength [118,119].

The sclera is a dense connective tissue of ≈24 mm in diameter, with a thickness ranging from 1 to 1.3 mm [120]. Its primary role is to provide mechanical support and protection for the retina and other delicate intraocular structures, while displaying nonlinear viscoelastic behaviour that absorbs and dissipates dynamic loads [118,119,121,122]. Structurally, the sclera is dominated by collagen—approximately 80% of its dry weight—which supplies tensile strength, while a minor elastin fraction (≈2% dry weight) confers elastic deformability and recoil [119,123,124]. Additionally, the ECM contains interfibrillar proteoglycans and glycoproteins that regulate hydration and thereby modulate the tissue's mechanical response [125]. Together, these components create a robust yet adaptable tissue capable of withstanding mechanical stresses while maintaining ocular integrity.

3. Optical methods

Naked eye visualisation is a widely used qualitative technique for assessing object deformations, but alternatives are needed to avoid subjective errors. While this review focuses primarily on techniques for observing the superficial deformation of the eye and the skin, optical methods have also enabled imaging within tissues. Such methods are essential in drug delivery applications to visualise how the injectate perfuses the tissue after administration. A novel approach involves staining tissues with highly absorbent molecules, such as food colouring, to enhance internal visualisation [126]; a comprehensive review can also be found elsewhere [127].

This section is organised into three parts: Section 3.1 (Optical imaging) describes optical readout methods that quantify geometry and displacement in experiments where the loading is not prescribed by the measurement technique; for example high-speed imaging, digital image correlation, and interferometric approaches. Section 3.2 (Optical elastography) covers methods that pair optical readout with a controlled excitation (mechanical, acoustic, or optical) to estimate mechanical parameters e.g., Brillouin microscopy, optical coherence elastography, and active speckle/phase methods. Section 3.3 (Summary of optical measurement techniques) integrates characteristic spatial and temporal resolutions, typical interrogation areas, and other practical considerations for operational planning.

3.1. Optical imaging

Optical imaging comprises readout-only techniques that visualise and quantify geometry, displacement and strain without prescribing the mechanical load. In this review, they are treated as visualisation methods in their own right; nevertheless, they often serve as the sensing arm of elastography experiments: when paired with a controlled external excitation (e.g., indentation, suction/bulge, air-puff, vibration or acoustic radiation force), they enable estimation of mechanical parameters. Multiple optical readouts can also be combined (for example, high-speed (HS) imaging with laser profilometry (LP), or optical coherence tomography (OCT) with phase-sensitive tracking) to meet the spatiotemporal demands of a given protocol.

The optical techniques reviewed here directly provide kinematic observables, such as surface topography, displacement and strain fields, or internal motion/wave speeds. However, they do not directly measure stress or constitutive constants. Mechanical properties are therefore obtained indirectly with e.g., a controlled mechanical excitation or boundary condition that is applied such as indentation, pressure, impulse/impact, harmonic excitation, etc. The resulting spatio-temporal displacement field is recorded optically, and an appropriate mechanical model or inverse finite-element procedure is used to fit the measured response. Among the parameters that can be extracted we have elastic modulus, surface tension or viscoelastic coefficients.

Several powerful optical techniques provide structural or compositional contrast that correlates with biomechanics and have been applied to skin and ocular tissues. Examples include two-photon tomography/SHG (sub-micrometre imaging of collagen/elastin architecture) [128,129] and Raman spectroscopy/microscopy (molecular composition, hydration, cross-linking) [130,131]. While highly informative for ECM organisation, these methods fall outside the ≈1–100 μm spatial window and sub-millisecond temporal regime emphasised in this review. Likewise, X-ray scattering/diffraction maps collagen orientation over wide fields of view but is generally not suited to fast, *in vivo* transient loading at these mesoscopic scales [132]. Accordingly, we focus on optical imaging readouts and optical elastography approaches that quantify mechanical response under controlled external stimuli within our target spatiotemporal window.

This section provides an overview of different methods, introducing their operating principles, and their respective spatial and temporal working ranges.

3.1.1. High-speed (HS) imaging main features

HS cameras are instrumental in capturing and supporting the analysis of events unfolding in the range from milliseconds to nanoseconds, spanning various scientific domains like fluid dynamics and biomechanics. To capture rapid phenomena with high spatial resolution, the camera needs to be coupled with the appropriate lenses and optical tools. Short exposure times, often on the order of microseconds or nanoseconds, are required to avoid motion blur. The exposure time must satisfy $\tau \leq r_p/(Mu)$, where r_p is the sensor pixel size, M is the optical magnification, and u is the typical object velocity [133]. The frame rate should exceed twice the driving frequency of the phenomenon being studied, with the optimum frame rate estimated using $f = Nu/l$, where N is the required number of samples, and l is a typical length scale [86].

Optical magnification (M) must satisfy $M > 2r_p/R_s$, where R_s is the spatial resolution defined by the Rayleigh criterion [134]. Higher magnification increases spatial sampling density but narrows the field of view and reduces image brightness. Ultimately, the achievable spatial resolution is determined mainly by the sensor pixel size (and the optical diffraction limit), which sets the smallest feature that can be reliably resolved. Cameras with larger pixels are limited to capturing coarser details, while those with smaller pixel sizes enable finer spatial resolution, allowing the visualisation of sub-micrometre structures (see Table 2). Temporal resolution, on the other hand, depends on the system's maximum frame rate, which can range from a few frames per second (fps) in standard video systems to several million fps in specialised high-speed setups. However, achieving extremely high frame rates typically

involves trade-offs, such as reduced spatial resolution or image size (see Table 2).

HS imaging limitations. A major limitation is the inverse relationship between frame rate and spatial resolution: as frame rates increase, the number of recorded pixels decreases. For example, a basic high-speed CMOS camera achieves 35,000 fps with a resolution of 100×100 pixels, while high-end models can reach up to 225,000 fps [86]. Optical resolution is fundamentally constrained by diffraction: in far-field imaging the smallest resolvable feature scales with the wavelength and inversely with the numerical aperture (commonly approximated as $\sim\lambda/(2,\mathrm{NA})$), which sets a practical limit on the finest details that can be distinguished [135]. Temporal sampling constraints can also introduce aliasing: if the motion frequency of interest exceeds half the frame rate (the Nyquist limit), fast oscillations or rapid deformations will appear as slower, misleading artifacts in the recorded sequence. Similarly, spatial aliasing may occur when object displacements between frames exceed one pixel, resulting in moiré-like distortions or ambiguities in tracking [136,137].

Another limitation is that HS imaging typically produces 2D images, requiring multiple cameras or advanced techniques to construct 3D representations. Furthermore, achieving well-exposed images at short exposure times requires intense lighting, which can be challenging in certain experimental setups, e.g., when heating a biological sample can lead to water evaporation or changes in the mechanical properties of the sample. These constraints must be carefully considered when designing HS imaging experiments.

HS imaging in soft and biological materials. HS imaging has been widely applied in fluid mechanics, medicine, and biomechanics, often combined with other techniques for *in vitro*, *ex vivo* and *in vivo* studies. In fluid mechanics, HS imaging has been coupled with shadowgraphy [138], Schlieren imaging [139], particle tracking velocimetry [140], particle imaging velocimetry [141], and holographic or interferometric imaging [142,143]. These combinations enable detailed analysis of rapid fluid dynamics phenomena.

In medical applications, HS imaging has been used to assess needle and liquid jet injections, providing insights into injection dynamics and tissue response [13,20,146,147], see Fig. 4(a). For example, HS polarisation cameras have enabled the study of the temporal evolution of cavity shape, stress intensity fields, and stress vector fields during microjet injections into gels [144] (see Fig. 4(b)). This technique, based on polarised light transmission through birefringent objects, enables visualisation of dynamic stress and strain fields [148,149]. The latter are concrete examples of HS imaging applied to drug delivery applications.

In biomechanics, HS imaging has been applied to study the dynamics of sheep eye stretching, providing real-time insights into tissue deformation and stress distribution [150]. These examples demonstrate the versatility of HS imaging in studying rapid phenomena in soft and biological materials, making it a powerful tool for both fundamental research and practical applications.

3.1.2. Event cameras (EC) main features

ECs, also known as neuromorphic cameras or dynamic vision sensors (DVS), represent a novel imaging technology that operates asynchronously, emulating the efficiency of biological vision systems by responding exclusively to changes in brightness within a scene [151,152], see Table 2.

ECs differ fundamentally from conventional and HS cameras by generating events only where there is a change in pixel intensity, rather than capturing frames at fixed intervals. This asynchronous operation enables microsecond temporal resolution and low-latency, high-speed responsiveness, making them ideal for real-time processing applications [153].

A key advantage of EC is their high dynamic range of 120 dB, which is comparable to that of the human eye. This allows them to capture scenes with both bright and dark elements simultaneously without sensor saturation [151,153]. Dynamic range refers to the ratio between the brightest and darkest tones that can be captured in an image, measured in decibels (dB) due to the logarithmic scale used to express large intensity ratios. The asynchronous nature of event cameras also contributes to lower power consumption compared to traditional cameras, as information is transmitted only during dynamic events [154]. ECs offer a cost advantage over traditional high-speed cameras, with commercial prices approximately one-fifth of those of high-speed cameras, making them an attractive option for researchers.

EC imaging limitations. As a relatively new technology, there is a limited body of research employing event cameras for measuring the mechanical properties of biological tissues. Their application in biomechanics and soft materials is still in its early stages, and further development is needed to fully exploit their potential. Additionally, while ECs offer high temporal resolution, their spatial resolution dramatically increases event rates, straining data bandwidth and processing pipelines [155,156]. This often leads to slow readouts and bus saturation, which in turn cause timestamp errors and dropped events, potentially limiting their ability to capture fine details in some applications [154,157].

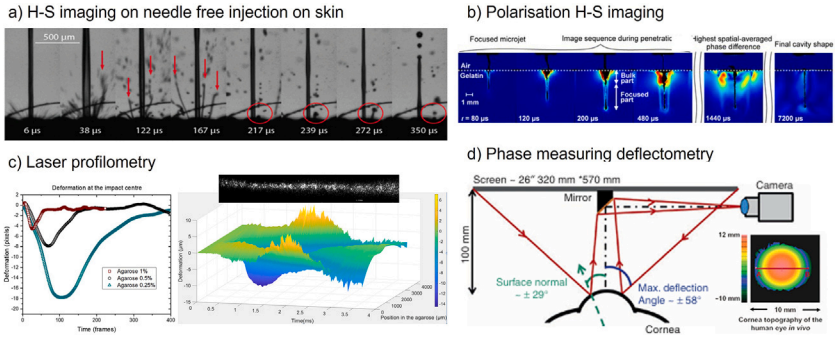


Fig. 4. Optical methods for visualising mechanical interactions in soft materials. (a) Visualisation with HS imaging on the needle-free injection on the skin [Reprinted from reference [20], which is licensed under CC BY 4.0]. (b) Polarisation high-speed imaging capturing stress distribution and cavity dynamics during microjet injection into a hydrogel (time range: 80–7200 µs) [Reprinted from reference [144], which is licensed under CC BY 4.0]. (c) Laser profilometry measurements of agarose deformation (d) Schematic of phase-measuring deflectometry (PMD) used to retrieve corneal geometry; inset: example corneal elevation/topography [Reprinted from reference [145], which is licensed under CC BY 4.0].

Another challenge is the need for specialised algorithms and processing techniques to interpret the asynchronous event data, which differs significantly from the frame-based (more intuitive) data produced by conventional cameras. This can increase the complexity of data analysis and require additional computational resources.

EC imaging in soft and biological materials. Event cameras have demonstrated promising applications in various fields, including the measurement of mechanical properties and dynamic behaviour of materials. For example, EC coupled with stroboscopic lighting has been used to measure the deformation of an acrylic sheet and the vibration of a polyurethane elastomer at 1000 fps [158]. These experiments highlight the potential of event cameras for capturing rapid mechanical phenomena with high temporal resolution.

ECs have also been applied to 3D object scanning and the generation of high-speed videos at approximately 5000 fps [159–162]. These capabilities make event cameras a promising tool for studying dynamic processes in soft and biological materials, such as tissue deformation or fluid-structure interactions.

Although the use of event cameras in biomechanics is still emerging, their unique features and cost-effectiveness suggest significant potential for future applications in measuring the mechanical properties of biological tissues and other soft materials, contributing to a better understanding of drug delivery mechanisms.

3.1.3. Laser profilometry (LP) main features

LP is a non-destructive, non-contact optical measurement technique used for obtaining precise and detailed surface profiles of objects. By projecting a laser beam onto a surface and capturing the reflected light, this method enables the reconstruction of 3D surface topography.

LP relies on the principle of triangulation, where the angle of the reflected laser beam is used to calculate the distance between the laser source and the surface to be measured. This technique provides highly accurate non-contact surface measurements, with the accuracy depending on the laser beam thickness, which can be as small as 1 μm. Typical analysis areas for laser profilometry are 10 mm × 10 mm [163].

When combined with an HS camera, laser profilometry achieves a typical depth resolution of 0.1 mm, enabling real-time measurement of surface features [164,165]. The temporal resolution of the system is limited by the acquisition rate of the HS camera or other signal acquisition systems. This combination of high spatial and temporal resolution makes laser profilometry a powerful tool for studying dynamic surface deformations in the range of micrometres and milliseconds.

LP imaging limitations. A key requirement is that the surface being measured must be reflective. For transparent or non-reflective surfaces, a thin material layer, such as gold, must be deposited to assist measurements [166]. This additional step can complicate the experimental setup and may not be feasible for all applications.

While LP can achieve high spatial resolution, its ability to capture rapid deformations, and temporal resolution, is constrained by the frame rate of the HS camera or other signal acquisition systems like the ECs. Additionally, the technique is sensitive to environmental factors such as vibrations ambient light, humidity, and other conditions which can affect measurement accuracy.

LP imaging in soft and biological materials. While LP has long been used to assess the static topography of biological materials such as skin and the eye [167,168], recent advances demonstrate that LP can also perform direct, time-resolved measurements of surface deformation. For example, a 2023 patent application from our group proposes an experimental configuration in which a microfluidic jet with known impact velocity deforms a gel or skin sample; the out-of-plane surface displacement along a line normal to the impact is tracked directly by LP (in combination with a high-speed camera), with reported sensitivity to displacements on the order of 10 μm at rates up to 10 kHz [165] (see Fig. 4(c)). The maximum displacement following the impact can be correlated to the elastic (storage) modulus, while the spatial and temporal evolution of propagating waves and their attenuation provide information on the loss modulus—i.e., LP provides a direct optical measure of deformation that can be quantitatively related to both elastic and dissipative mechanical properties.

The combination of laser profilometry with HS imaging offers a powerful approach for real-time, high-resolution surface deformation analysis. While its application to characterise biological materials is still in its early stages, its non-destructive and non-contact nature makes it a promising tool for future research in biomechanics and soft material characterisation, which could improve the drug delivery.

3.1.4. Structured light profilometry (SLP) main features

SLP is a non-contact optical technique that projects known patterns (e.g., grids, stripes, or random points) onto a surface and captures the resulting deformations with one or more cameras [169,170]. By analysing these distortions—either through triangulation in a stereo-vision setup or by measuring modulation delays in an active illumination scheme—SLP reconstructs high-resolution 3D surface maps without physical contact [171,172].

Pattern strategies in SLP range from simple binary codes and statistical dot patterns to phase-based sinusoidal fringes, each offering different trade-offs between spatial resolution, measurement speed, and noise robustness [173–175].

Recent advances in SLP include the integration of neural networks and deep learning to enhance the accuracy and speed of 3D reconstruction. Innovations in pattern generation and projection have further improved this technique [176–180].

SLP limitations. The technique requires a clear line of sight between the projector, the cameras, and the object under study, which can be challenging for complex or highly curved surfaces. Moreover, the accuracy of SLP depends on the quality of the projected patterns and the calibration of the system, which can be sensitive to environmental factors such as ambient light and vibrations.

For high-speed applications, the rapid switching of patterns in binary coding or sinusoidal phase encoding can introduce challenges in synchronisation and data processing. Furthermore, the computational complexity of reconstructing 3D surfaces from captured images can be significant, particularly when using advanced techniques such as Fourier transforms or neural networks.

SLP imaging in soft and biological materials. SLP has been applied to a variety of soft and biological materials, demonstrating its versatility and precision. In skin characterisation, sinusoidal fringe projection has been used for hand palm print recognition, providing detailed surface topography for biometric applications [181]. Gradient illumination techniques have also been employed for skin imaging in dermatology, enabling non-invasive analysis of skin texture and morphology [182].

In biomechanics, SLP has been used to measure dynamic deformations, such as the vibration of membranes and the flapping motion of bio-inspired robots. For example, sinusoidal phase encoding has achieved high-speed measurements of membrane vibrations at 1000 fps with sub-millimetre accuracy [183], while binary coding has been used to analyse the rapid flapping motion of robotic wings at 5000 fps [184]. These applications highlight the potential of SLP for studying dynamic mechanical behaviour in soft and biological materials.

The integration of SLP with advanced computational techniques, such as neural networks and deep learning, opens new possibilities for real-time, high-resolution 3D reconstruction in biomedical research and clinical applications, such as monitoring drug delivery [175,185]. Although SLP has traditionally been applied to surface topography, it can also record time-resolved, full-field displacement and strain maps during transient loading. These experimentally measured deformation

fields can be fed into inverse methods or fitted with constitutive models to extract quantitative mechanical parameters e.g., Young's modulus, shear modulus, and viscoelastic relaxation times. These features make SLP a versatile tool not only for geometry mapping but also for mechanical characterisation of soft and biological materials. As electronics, data processing and computational power continue to improve, SLP is expected to play an increasingly important role in this area.

3.1.5. Electronic speckle interferometry (ESPI) and digital holography (DH) main features

ESPI is an optical technique to measure microscopic deformations and displacements in objects. ESPI relies on the interference of coherent light waves generated by lasers [186,187]. When the coherent light waves interact with a rough surface, they create a random interference pattern known as a speckle pattern [188].

In ESPI, the reference beam is directed to a mirror or a reference surface, while an object beam illuminates the object under investigation. Both beams create speckle patterns on the surface of the object, and their overlap produces an interference pattern recorded by a camera. These interference patterns provide surface information without requiring calibration, as the fringe spacing depends on the laser wavelength [188,189]. For a more in-depth understanding of the optical principles behind speckle interferometry, we direct the reader elsewhere [190].

ESPI has high sensitivity, capable of measuring deformations at the micron or sub-micron level. This enables the measurement of mechanical properties of inhomogeneous and anisotropic materials, such as elastic modulus, shear modulus, and Poisson's ratio [191–193]. Additionally, it can be used in both reflective and translucent materials [194]. The technique allows for tracking stress and strain distributions and often employs phase-shifting techniques, which systematically alter the phase of one of the beams for more precise measurements [195,196].

DH operates based on a principle similar to ESPI but introduces a phase difference between the reference beam and the object [197]. This modification enables dynamic measurements and has been effectively combined with high-speed cameras to capture dynamic deformations in materials [142].

ESPI and DH limitations. Both techniques are highly sensitive to environmental vibrations, which can introduce noise and affect measurement accuracy. Therefore, vibration isolation setups are often required. Additionally, speckle contrast and resolution depend on the optical setup and the coherence of the laser source, which may impose restrictions on certain applications. For example, *in vivo* biomedical imaging, such as monitoring blood flow in tissues or assessing wound healing, patient motion or physiological activity can degrade image quality. DH, while offering enhanced phase contrast, similarly requires stable environmental conditions to achieve high-quality reconstructions.

ESPI and DH in soft and biological materials. Beyond readout-only operation, ESPI/DH can also serve as the sensing probe of elastography: when coupled to a controlled excitation (e.g., vibration/oscillation, driven surface waves or pressure-driven loading), they can be operated as laser speckle elastography or holography-based elastography to derive mechanical response [198]. One of the advantages of ESPI is its versatility, allowing deformation testing in both two-dimensional planes and full three-dimensional reconstructions of objects [196,199]. The technique has been successfully used to measure deformation in biological membranes and soft tissues under biaxial stretching and compressive pressure [188,194,200,201]. A summary of studies using speckle interferometry for mechanical characterisation of biological tissues can be found elsewhere [202].

Digital holography has been used to assess the rigidity of *ex vivo* porcine skin under varying exposure times to UV light, revealing an increase in skin rigidity following UV light exposure [203].

3.1.6. Phase measuring deflectometry (PMD) main features

PMD is an optical measurement technique designed for the accurate measurement of surface shapes, particularly beneficial for large and reflective surfaces [204]. This method involves projecting fringe patterns of UV or visible light onto the surface of an object, creating variations in brightness that deform based on the local curvature of the object [205]. By measuring the phase shift of these deformed fringe patterns, PMD captures detailed information about the object's surface [205]. The technique is highly sensitive to local slopes, allowing it to measure surfaces with significant variations in topography [206]. A typical PMD setup includes an LCD screen connected to a computer to generate the phase-shifting sinusoidal fringe patterns, which are captured upon reflection from the surface by a camera [207].

Accurate calibration is essential in PMD, which involves imaging a known reference surface to establish the relationship between the measured phase and actual surface roughness [204,206,207]. PMD is particularly well-suited for measuring specular or shiny objects and free-form surfaces. Surface deflectometry can measure heights of 200 ± 40 μm, with a lateral resolution of ≈100 μm; see Table 2.

Although PMD does not directly measure internal stresses, these surface displacement fields are mechanically informative: when a known mechanical stimulus (for example a controlled indentation, suction pulse, air-impulse, or pressure change) is applied, the resulting surface height/slope changes measured by PMD can be converted into local strain fields. Additionally, this information can be used to estimate effective mechanical parameters, e.g., apparent stiffness or spatial variations in compliance over the interrogated patch using forward or inverse elasticity models.

PMD limitations. PMD faces challenges related to environmental conditions, particularly variations in ambient light, which can affect measurement accuracy. Additionally, the technique is primarily designed for highly reflective surfaces, limiting its direct application to diffuse or absorptive materials without modifications [208]. Calibration procedures can also be complex, requiring a precise reference surface to ensure accuracy [206,207].

PMD in soft and biological materials. Recent developments in PMD include the use of two LCD screens [206,209,210], the addition of a projector to the measurement system [211], and the use of infrared light [212]. The first two variations of PMD allow the surface height profile to be directly reconstructed from the absolute phase, enabling accurate profiling of discontinuous surfaces [208]. Meanwhile, using infrared light minimises the impact of ambient light on measurement accuracy.

PMD holds promise for broader use based on attempts to integrate it with commercial mobile devices [213,214]; a rapidly advancing technology [208]. Consequently, it could enable *in vivo* skin and eye topography, as well as deformation measurements, using nothing more than a phone.

Single-shot deflectometry, which uses a single cross-sinusoid pattern rather than capturing a sequence of unidirectional sinusoidal patterns, has been successfully applied to measure the shape of the human cornea *in vivo* and track eye movements [145,215,216] (see Fig. 4(d)).

3.1.7. Optical coherence tomography (OCT) main features

OCT is a non-invasive imaging technique that uses interferometry to generate high-resolution, cross-sectional 3D images of highly scattering materials, such as most biological tissues. A conventional OCT system employs a Michelson interferometer to measure light echoes from tissues. Due to the smaller wavelength of light compared to sound waves in ultrasound (US) imaging, OCT achieves significantly higher spatial resolutions, typically in the range of 1–15 μm. Imaging depths depend on the scattering properties of the sample but are generally within 1–3 mm [217].

Traditional OCT systems consist of an interferometer with a low-coherence-length light source, which is split into a reference and a sample arm by a beam splitter. When a sample is illuminated, microscopic differences in its refractive index scatter the light, creating interference patterns. Step-wise analysis of reflections along the optical axis provides depth-resolved structural information. A full 3D cross-sectional image is obtained by scanning the beam (or sample) in the X and Y directions.

OCT limitations. While OCT offers high spatial resolution and is non-invasive, its imaging depth is limited to a few millimetres due to light scattering in highly opaque tissues [218]. Additionally, OCT systems require precise optical alignment and are sensitive to motion artifacts, making *in vivo* imaging challenging without stabilisation techniques. Accurate interpretation of OCT images requires expertise, since variations in tissue scattering can obscure anatomical boundaries and create ambiguity when distinguishing structures [219].

OCT in soft and biological materials. OCT has been extensively used in ophthalmology for diagnosing retinal diseases, assessing corneal structures, and monitoring disease progression. In dermatology, OCT has also become an important non-invasive imaging modality because it provides real-time cross-sectional views of skin architecture to depths of roughly 1–2 mm, allowing detailed visualisation of the epidermis and superficial dermis [220]. Clinically, OCT supports the assessment and differential diagnosis of a variety of conditions—for example, non-melanoma skin cancer (especially basal cell carcinoma), inflammatory dermatoses, onychomycosis and infectious/parasitic infestations [221]. Beyond diagnosis, OCT and its functional extensions (OCT angiography, dynamic-contrast OCT) can resolve cutaneous microvasculature and vascular dynamics, which is useful for studying angiogenesis, inflammatory blood-flow changes and treatment response [222,223]. OCT is also widely used in research and development of transdermal drug delivery: high-resolution OCT visualises microneedle penetration, in-skin dissolution of dissolving microneedles and real-time drug–tissue interactions, making it a standard tool when testing microneedle designs and topical formulations [224,225]. Finally, recent high-resolution implementations such as line-field confocal OCT and full-field OCT increase cellular-level detail and enable faster, volumetric, or dynamic imaging—features that make OCT increasingly valuable for both clinical dermatology and for mechanistic studies of skin deformation and short-timescale dynamics (e.g., following microneedle injection) [226,227].

3.2. Optical elastography

Optical elastography comprises techniques that were developed to directly measure the mechanical properties of tissues and materials. In contrast to purely visualisation-based approaches, these methods measure how a sample deforms or supports mechanical waves under a known or induced load, enabling the estimation of stiffness, viscoelastic parameters, or wave propagation speeds [228,229]. In this section, we review representative modalities spanning distinct contrast mechanisms and spatiotemporal operating regimes: digital image correlation (DIC) for surface strain mapping under macroscopic loading [230]; optical coherence elastography (OCE) for phase- and displacement-resolved assessment of subsurface deformations and guided-wave dynamics [231]; laser Doppler vibrometry (LDV) for non-contact measurement of out-of-plane surface velocity and frequency response [232]; and Brillouin microscopy, which probes spontaneous Brillouin scattering from thermally excited acoustic phonons to map the GHz-scale longitudinal modulus at sub-micrometre resolution without external loading [233]. Although these techniques differ in penetration depth, sensitivity, and temporal bandwidth, they share the central principle of linking an optical observable to a mechanical quantity through an explicit mechanical model.

3.2.1. Digital image correlation (DIC) main features

DIC is a technique for quantitatively analysing the deformation and strain distribution of materials under mechanical loading. A reference pattern is applied to the surface using paint or small particles before deformation. By comparing a series of images captured before and after deformation, DIC enables the calculation of displacement, strain, and deformation gradients with sub-pixel resolution and accuracy [230,234].

The DIC process begins with capturing a series of images of the object under different loading conditions, typically using a high-resolution camera. To enhance detection accuracy, the material surface must exhibit a random high-contrast pattern with an approximately equal distribution of black and white regions. The images are then divided into subsets of a few pixels in width, often in square or rectangular regions, which serve as reference points for tracking. The displacement of these subsets is determined by comparing their original positions in the reference image with their new positions in the deformed image through cross-correlation techniques [235]. This analysis provides displacement vectors in both horizontal and vertical directions. From this data, strain is computed as a measure of deformation [234]. Optimal speckle dot sizes are 3–5 pixels; with typical microscope objectives producing pixel sizes between 0.1 and 10 μm, corresponding to feature sizes of roughly 0.3–50 μm. DIC can achieve sub-pixel displacement accuracies down to 0.01 pixels—equating to approximately 1–100 nm, depending on the imaging magnification [230]. There are open-source libraries, such as those for Matlab or C++, that provide algorithms to implement DIC and analyse images on a personal computer [234,236].

DIC limitations. Applying DIC to biological tissues presents challenges due to their natural hydration, which can distort the speckle pattern through moisture-induced spreading and reflections. To address this, researchers often use a mist of fast-drying enamel paint or temporary tattooing for skin applications [185,237]. For transparent tissues, such as the eye, a non-reflective white background may be necessary to generate an adequate pattern [185]. In cases where the specimen exhibits an intrinsic random texture or inhomogeneity, these natural patterns can be directly exploited for DIC [238]. Additional techniques, such as immersing soft samples in methylene blue followed by applying a white speckle pattern, have also been employed [239–241].

DIC imaging in soft and biological materials. DIC has been extensively applied to study the mechanical response of human skin under point loading, with data integrated into finite element models to estimate hyperelastic parameters [242]. Similarly, DIC-measured quantities have been used in specimen-specific finite element models of the sclera [243]. More recently, *in vivo* suction tests combined with DIC have been employed to characterise skin anisotropy over an area of 80 mm^2 with a depth resolution of 1 μm [244].

DIC has been applied in high-speed, *in vivo* experiments to extract modal (natural or resonant) frequencies of skin by tracking skin motion with temporal resolutions on the order of 1 ms and displacement sensitivity lower than 10 μm over areas of ≈1500 mm^2 [245]. For example, Panchal et al. [246] used DIC together with finite-element modal analysis and reported clear resonant peaks at ≈145.6–156.1 Hz in forearm measurements (different test cases), demonstrating agreement between DIC and finite-element modal predictions. Reported resonant frequencies in the literature vary with anatomical site and tissue condition (normal skin shows dominant peaks near ≈50, ≈100 and ≈150 Hz, whereas scarred/fibrotic tissue can exhibit higher peaks up to ≈220–260 Hz), which explains the broader range cited (≈106–240 Hz) as reflecting different sites/conditions and measurement methods [247]. Additionally, DIC has been employed in conjunction with high-speed imaging at 40,000 fps to track PDMS and agarose gel deformations with a spatial resolution of 115 μm [165,248], as a proxy to understand skin deformation under needle-free drug delivery (see Fig. 5(a)).

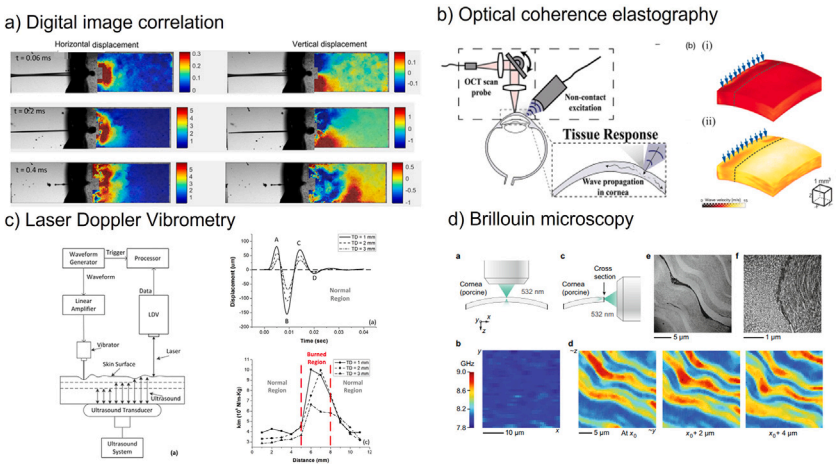


Fig. 5. (a) Shear rate analysis in a hydrogel under jet impact, showing horizontal and vertical displacement profiles over time using DIC [Image adapted from [165]]. (b) OCE schematic and group-velocity maps of the cornea at intraocular pressures (IOP) of 10 and 40 mmHg, illustrating IOP-dependent stiffness [Reprinted from reference [249], which is under CC BY 4.0]. (c) Block diagram of the LDV-based setup used to generate, detect and monitor surface waves on skin, with a representative impulse response from a porcine sample (burned region indicated) [Reprinted by permission from reference [250], Copyright 2014, Elsevier]. (d) High-resolution Brillouin imaging of *ex vivo* porcine cornea (532 nm): (i) en-face imaging schematic (spot ≈0.5 μm); (ii) en-face Brillouin map (uniform values); (iii) cross-sectional schematic (laser focused below surface); (iv) depth-resolved Brillouin maps showing lamellar structure; (v–vi) TEM micrographs confirming lamellae and aligned collagen fibrils (keratocyte visible) [Reprinted from reference [251], which is licensed under CC BY 4.0].

3.2.2. Optical coherence elastography (OCE) main features

Optical coherence elastography is an OCT–based family of methods that measures tissue displacements produced by a controlled excitation and converts those displacements (or their spatial/temporal derivatives) into local mechanical information such as strain or derived moduli [231]. OCE implementations exploit amplitude and phase sensitive OCT to achieve nanometre scale displacement sensitivity and high frame rates. They generate transient or steady mechanical waves using contact indentation or vibration, air puff, photothermal or optical excitation, and ultrasound acoustic radiation force. OCT then records wave propagation or the local tissue response [55,252]. Because of the high spatial resolution of OCT and the sensitivity of phase measurements, OCE can probe mechanical dynamics on the sub-millimetre length scale and on the sub-millisecond to millisecond timescales relevant to the present review [249,253]. OCE inherits the penetration depth of OCT in scattering tissues (typically up to ~1–2 mm in most biological tissues under common wavelengths) [231].

Several OCE submodalities have been reported, defined by the excitation mechanism and the resulting mechanical wavefield. For the scope of this work, the most relevant are:

Surface Acoustic Wave OCE (SAW-OCE) launches guided surface/Lamb waves, often via impulsive or air-puff stimuli, and allows stiffness estimations from their phase/group-velocity dispersion [254]. Acoustic-Radiation-Force OCE (ARF-OCE) uses focused ultrasound to generate localised body forces while phase-resolved OCT tracks the induced motion [255]. Reverberant 3D OCE (Rev3D-OCE) recovers depth-resolved elasticity from diffuse or semi-reverberant wavefields without requiring clean single-mode tracking [256]. Lastly, Photothermal OCE (PT-OCE) relies on modulated optical absorption to drive thermoelastic stresses, enabling all-optical, co-axial excitation/detection with rich high-frequency content [257]. A more detailed overview and comparison of these and other OCE submodalities can be found elsewhere [228,229].

OCE limitations. Many implementations require precise control or calibration of the excitation (force amplitude, spatial distribution, frequency content). Contact excitations can perturb boundary conditions and non-contact excitations (air-puff, photonic) may have limited bandwidth or penetration, affecting repeatability and comparability between systems [249]. *In vivo* applications e.g., to characterise the eye and skin must cope with physiological motion (eye saccades, breathing, pulse) and environmental vibrations; as such phase-sensitive OCE is particularly sensitive to such noise and requires appropriate gating, averaging, or motion-compensation strategies [249]. Inter-system differences (OCT source, scanning scheme, excitation type, signal processing) complicate direct comparison of numerical mechanical values across studies; standard phantoms and cross-validation are still an active area of development [231].

OCE in soft and biological materials. OCE is increasingly applied in dermatology and cosmetology to map local skin mechanics, to quantify disease-related stiffness changes, anisotropy, and monitor hydration or topical-agent penetration [253,258]. Recent work demonstrates OCE's ability to detect changes in the mechanical response associated with substance penetration and skin hydration/dehydration in real time, offering a non-invasive alternative or complement to optical or spectroscopic penetration assays, such as drug delivery [253].

OCE has also been widely adopted in ophthalmic research to map the spatial distribution of corneal stiffness, to monitor changes produced by collagen cross-linking, and to investigate the biomechanics related to intraocular pressure and disease states such as keratoconus and glaucoma [55,249] (see Fig. 5(b)). SAW-OCE, static/compressive OCE (indentation with OCT readout), and air-puff OCE have all been used to extract displacement, strain and wave-speed metrics that correlate with tissue stiffness and pathological alterations [259]. The combination of micron-scale spatial resolution and millisecond temporal sampling is particularly valuable for resolving transient responses of the cornea and for comparing mechanical behaviour before/after clinical interventions [231].

ARF-OCE has mapped corneal elasticity, detected stiffening after collagen cross-linking, and probed procedure- and pressure-related corneal biomechanics [260–262]. Rev3D-OCE provides layer-specific corneal mechanics and depth-dependent stiffening profiles after cross-linking,

with recent extensions enabling multi-frequency characterisation of viscoelastic dispersion [256,263]. PT-OCE offers all-optical, co-axial excitation/detection with rich high-frequency content for probing superficial layers and microstructures [264].

3.2.3. Laser Doppler Vibrometry (LDV) main features

Laser Doppler Vibrometry is a non-contact optical technique that measures the frequency shift of back-scattered laser light from a vibrating surface, known as the Doppler shift. This frequency shift is proportional to the velocity of the surface motion and is detected using interferometers, such as a Mach-Zehnder or Michelson interferometer [232].

LDV is a versatile technique capable of both single-point and scanning measurements, enabling detailed mapping of vibration patterns on complex surfaces. The most common lasers used in LDV are He-Ne lasers with a wavelength of 632.8 nm, which are suitable for highly reflective objects, and infrared lasers with a wavelength of 1550 nm, which are preferred for rough or low-reflectivity surfaces, such as human skin [265].

LDV configurations vary depending on experimental needs. Single-point LDV focuses the laser beam on a single point, measuring vibration characteristics at that location, an approach for localised measurements. Scanning LDV moves the laser beam across the moving surface in patterns such as raster or zigzag, allowing vibration properties to be measured at multiple points and creating vibration maps of entire surfaces [232]. Multi-point LDV measures vibrations at multiple points simultaneously using multiple laser beams or detectors, providing a comprehensive view of surface dynamics [266]. This technique works best for periodic vibrations.

LDV offers high displacement resolution, typically on the order of 10 nm, with a frequency range extending up to 20 MHz [267]. The temporal resolution of LDV measurements is typically around 1 ms, making it suitable for capturing rapid dynamic events [250].

LDV limitations. The technique requires a clear line of sight between the laser and the target surface, which can be challenging for complex or irregular geometries. Additionally, the accuracy of LDV measurements can be affected by surface reflectivity, with highly reflective or transparent surfaces requiring specialised laser wavelengths or surface treatments.

Another limitation is the sensitivity of LDV to environmental vibrations and noise, which can interfere with measurements. For scanning LDV, the time required to scan an entire surface can limit its applicability to real-time or high-speed applications. Furthermore, the complexity of multi-point LDV systems can increase the cost and setup requirements, making them less accessible for some applications.

LDV imaging in soft and biological materials. LDV has been employed to determine the Young's modulus of hydrogels by exciting them with a piezoelectric element within the frequency range of 0 to 2.5 kHz [268]. This approach provides a non-contact method for characterising the mechanical properties of soft materials.

In dermatology, LDV has been used to measure facial expression activity and analyse skin surface waves. For instance, LDV was employed to assess the viscoelastic properties of porcine skin by generating surface waves with a ball-tipped device and subsequently measuring them [250] (see Fig. 5(c)). This technique offers a non-invasive way to study the mechanical behaviour of skin. Details on the method used to generate surface waves for mechanical probing can be found in Section 5.3.

In ophthalmology, LDV has been applied to measure intraocular pressure using a handheld vibrometer [269]. Additionally, analysing the normal vibrations of the eye with LDV has shown potential for determining its mechanical properties [270].

Ongoing research is focused on miniaturising LDV techniques, with the goal of developing handheld devices for *in vivo* measurement of skin properties [271]. This advancement could enable real-time, high-resolution measurements in clinical settings, such as (i) selecting formulations or penetration enhancers for drug delivery, (ii) detecting immediate mechanical responses following application that correlate with uptake, and (iii) guiding localised administration in personalised treatment scenarios, further expanding the utility of LDV in biomedical research and diagnostics.

3.2.4. Brillouin microscopy (BM) main features

Unlike load-based elastography, BM does not apply a macroscopic external load. This technique relies on Brillouin light scattering, which occurs when light interacts with thermally excited longitudinal acoustic (elastic) waves (phonons) in the material, producing GHz-scale frequency shifts that are related to the local longitudinal elastic modulus [233]. This type of scattering is influenced by the compression of the material and its stiffness, as well as its viscoelastic properties [233]. The scattering is inelastic because photons either gain or lose energy, producing a measurable shift in optical frequency; in Brillouin scattering this shift typically lies in the gigahertz (GHz) range [272]. This frequency shift is challenging to detect with conventional spectrometers. Consequently, BM setups typically employ a virtually imaged phased array (VIPA) spectrometer, which offers the ultra-high spectral resolution needed to resolve the gigahertz-scale frequency shifts [272]; see Table 2.

Specifically, this frequency shift can be correlated with the longitudinal storage and loss moduli, which measure how a material responds to stress and strain when confined. Unlike Young's modulus and shear modulus, which gauge responses under different conditions (e.g., compression, shear, or elongation), the longitudinal modulus is part of the overall elasticity of a material, composed of multiple factors. Brillouin light scattering experiments can elucidate the entire elastic tensor of a material by altering the scattering angle [233,273].

Brillouin spectrometers have been integrated with complementary modalities such as Raman [274] and fluorescence imaging [275] to relate local elastic measurements to morphological, structural and biochemical features [233]. The Brillouin signal typically yields an axial spatial resolution (probing depth) on the order of 1–5 μm within the sample [233]. The effective temporal sensitivity of Brillouin measurements is governed by the material's relaxation dynamics: very fast relaxations (e.g., water at room temperature) occur on picosecond scales, slower viscous responses can lie in the millisecond range (e.g., highly viscous liquids), and extremely slow relaxations (e.g., aggregated protein structures such as amyloid plaques) may extend to seconds, all of which influence the Brillouin linewidth and shift [233,276].

BM limitations. To capture Brillouin signals, light must be gathered for periods ranging from milliseconds to seconds for each section of the sample being studied. This means that the recorded data represent an average over this time frame. Since it takes time to gather data from each point, creating a complete map of the material's properties can take anywhere from minutes to hours [233].

When sound waves propagate through a material, the mechanical response of the medium depends on the frequency of the waves relative to the material's intrinsic relaxation time. If the frequency is low (i.e., the oscillation is slow compared with the material's characteristic relaxation time), the material has sufficient time to undergo molecular rearrangements and dissipate stress, behaving as a viscous fluid, and thus, it behaves more compliantly or flexibly. In contrast, at high frequencies (for oscillations with periods negligible compared to the relaxation time), the material cannot reorganize quickly enough, so it resists deformation and behaves more like an elastic solid, appearing stiffer. When the wave period is comparable to the relaxation time,

partial relaxation occurs, leading to significant energy dissipation. This energy loss manifests itself as a broadening of the Brillouin spectral line, providing insight into the viscoelastic properties of the medium. Consequently, accurate Brillouin measurements of elasticity—especially under varying temperature and hydration—are more sensitive to these conditions than other techniques reported in this review, because the relaxation dynamics that govern the viscoelastic response (and thus the spectral linewidth and shift) depend critically on molecular mobility and water content within the sample [233].

BM in biological and soft materials. Brillouin spectroscopy has been reported in studying the elastic properties of *ex vivo* chicken skin, which had injuries induced by continuous wave laser irradiation [277]. The results indicated that the injured skin became softer than healthy skin. A similar method was used *ex vivo* to differentiate between melanoma and healthy tissues [278], based on the larger shift in the Brillouin peak observed in healthy skin compared to injured skin. *In vivo* measurements of human skin mechanical properties using BM and OCT showed that the Brillouin frequency shift in the epidermis was higher than in the dermis, corroborating the expected finding that the dermis is softer than the epidermis [279].

BM has been employed to measure the longitudinal modulus throughout the entire eye, and comparisons with Young's modulus obtained from compressive stress–strain tests have revealed a correlation between the two [276,280]. Additionally, another study assessed the microscopic anisotropy of the eye by visualising the elastic properties of interwoven fibre layers in an *ex vivo* porcine cornea [251] (see Fig. 5(d)).

3.3. Summary of optical measurement techniques

Fig. 6 and Table 2 summarise representative operating ranges for the optical visualisation and measurement modalities discussed in this review, and highlight the principal trade-offs relevant for dynamic mechanical testing. HS imaging provides excellent temporal resolution, from ~10^3 to ~10^6 fps depending on the sensor and ROI, at spatial resolutions limited by pixel size, optics and illumination. High frame rates are typically achieved at the expense of sensor area or spatial sampling and require powerful lighting for short exposures [86]. ECs deliver extremely low latency and microsecond-level time-stamping of brightness changes and are therefore advantageous for capturing very fast transients with minimal motion blur. However, their per-pixel spatial sampling and suitability for quantitative intensity-based measurements remain an active area of research and often require fusion with frame-based imaging for full scene reconstruction [152,160]. BM and other light-scattering mechanical spectroscopies provide label-free, sub-micrometre probing of viscoelastic contrasts at tens of ms to hundreds of ms per pixel, which limits real-time imaging of fast dynamics [273]. Interferometric techniques such as ESPI provide very high sensitivity to displacement and strain and can operate in dynamic modes to capture vibration and transient deformation. Their practical temporal bandwidth and full-field accuracy depend strongly on the optical configuration and the phase-demodulation method [190].

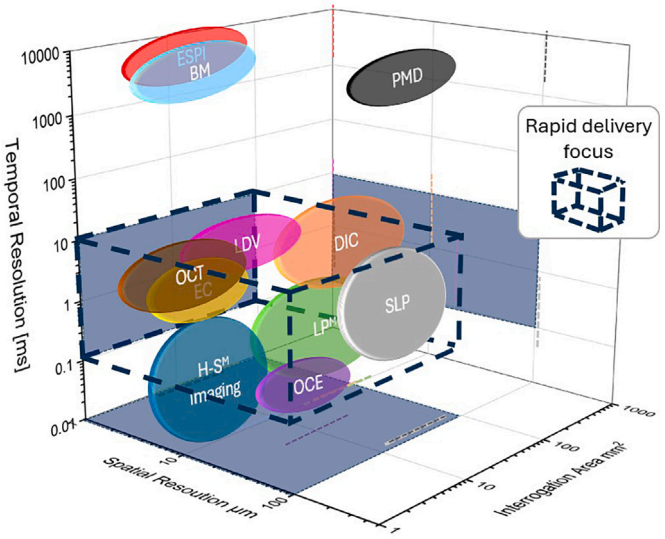


Fig. 6. Comparison of imaging and measurement techniques based on spatial resolution, temporal resolution and interrogation area. Each ellipsoid represents the typical performance range of a technique. Techniques such as high-speed cameras and event cameras offer excellent temporal resolution but are limited in spatial resolution and interrogation area. In contrast, Brillouin microscopy and ESPI provide high spatial resolution but operate at much slower acquisition speeds. This visualisation highlights the trade-offs among different modalities and helps guide the selection of appropriate tools for dynamic measurements in soft and biological materials. SR, IA and TR ranges correspond to peer-reviewed reports for each technique, unless otherwise indicated. Entries for which the reported range is taken primarily from manufacturer specifications are labelled "M"; further details are given in Table 2.

Table 2
Typical measurement capabilities of the optical techniques, spatial resolution (SR), interrogation area (IA), temporal resolution (TR) and some practical considerations, see Fig. 6.

Technique	SR (µm)	IA (mm^2)	TR (ms)	Considerations
HS camera^M (Section 3.1.1)	10	1	0.01–1	*Maximum TR is ≈5 ns [281] *Increasing the TR compromises SR and reduces the IA.
EC (Section 3.1.2)	1	10	0.2	*Maximum TR ≈1 µs *Fixed SR and TR.
LP^M (Section 3.1.3)	10	100	0.1	*Lateral SR depends on lens resolution. *Perpendicular/depth SR depends on the laser line thickness.
SLP (Section 3.1.4)	100	10	0.1	*SR and TR depend on the projected pattern.
ESPI (Section 3.1.5)	1	100	Static	*Not tested under dynamic deformations. *SR dependent on the laser wavelength and lens to create the speckle pattern.
PMD (Section 3.1.6)	100	10	Static	*Not tested under dynamic deformations
OCT (Section 3.1.7)	1	10	0.5	*Multiple layers of tissue can be scanned.
DIC (Section 3.2.1)	10	80	1	*Sub-pixel SR. *SR can be ca. 0.1 µm for *static* deformations.
OCE (Section 3.2.2)	15–100	10	0.1	*TR set by A-line rate/acquisition. *Trade-off with IA; *in vivo* requires motion compensation
LDV (Section 3.2.3)	0.01	10	1	*IA can be varied by using single-point, scanning or multi-point LDV.
BM (Section 3.2.4)	0.5–5	100	1×10^4–1×10^6	*Sensible to small changes in elasticity. *3D mapping without physical sectioning.

SR, IA and TR values correspond to typical literature demonstrations for each technique, unless otherwise indicated. Entries for which the reported range is taken primarily from manufacturer specifications are labelled "M".

4. Acoustic methods

Optical methods offer high-resolution imaging and significant contrast enhancement, achieving even higher contrast on reflective surfaces. For non-reflective materials, pretreatment is often required, limiting their application in biomedical imaging. Additionally, few optical techniques can image inside biological tissues to track internal structures during deformation. In contrast, acoustic methods, which have evolved significantly since the mid-20th century, provide deep tissue penetration, high biocompatibility, and real-time temporal resolution, making them indispensable for medical applications due to their affordability. Pioneered in the 1940s–1950s with pulse-echo ultrasound originally developed for submarine detection during WWII, acoustic imaging gained clinical traction through Ian Donald's groundbreaking obstetric ultrasound in the 1950s [282]. By the 1970s, B-mode ultrasound enabled real-time visualisation of soft tissues such as the liver and heart [282]. Subsequent refinements, such as Doppler ultrasound for blood flow analysis (1980s) and shear wave elastography to quantify tissue stiffness (1990s), expanded diagnostic capabilities in cardiology and oncology [283]. Since the 2000s, portable, low-cost ultrasound systems have democratised access to imaging in resource-limited settings [284–286].

4.1. Acoustic imaging

Acoustic imaging provides high-resolution structural and compositional maps that reveal tissue layers and heterogeneities. When combined with quantitative acoustic measurements or active excitation, it also yields direct estimates of mechanical properties [287]. Quantitative ultrasound and scanning acoustic microscopy measure acoustic parameters (speed of sound, attenuation, impedance) that can be used, with appropriate density and boundary-condition assumptions, to infer elastic moduli [288,289]. Because the mechanical response of soft tissues is frequency- and time-dependent, the millimetre spatial scale and the 0.1–100 ms deformation window emphasised in this review are particularly well matched to high-frequency ultrasound (HFUS) and Scanning Acoustic Microscopy (SAM) methods capable of resolving both rapid transient dynamics and localised mechanical heterogeneity. In practice, robust mechanical quantification requires careful calibration, model selection (elastic versus viscoelastic), and relating acoustic contrast to underlying composition and boundary conditions [290].

Although conventional ultrasound techniques are known for their low spatial resolution when compared to optical methods, recent advances have enabled the development of an ultrasonic nano-imaging system that can visualise a single cell [291]. Acoustic visualisation methods are generally divided into three main categories: Ultrasound Imaging (US), Acoustic Microscopy (AM), and Photoacoustic Imaging (PA).

4.1.1. Ultrasound (US) main features

Ultrasound waves are transmitted using a transducer (or an array of transducers) into the tissue. The acoustic (sound) waves encounter differences in acoustic impedances due to the variable density and bulk moduli of the medium. Some of these waves are backscattered or reflected into the transducer to be processed and combined to generate an image. US techniques, inherently involve a compromise between spatial resolution and penetration depth [292]. Higher frequencies (ω) lead to higher spatial resolution, but since attenuation increases with frequency, the penetration depth δ_p is reduced, i.e., $\delta_p(\omega) \propto 1/\omega^{\alpha}$ (with α a positive real number) [293–296].

US is commonly defined as sound waves with frequencies larger than 20 kHz. However, conventional medical equipment operates at frequencies ranging from 2 to 15 MHz, allowing imaging of large interrogation areas (see Table 3). Due to its low spatial lateral resolution (0.3–3 mm) [297], it is commonly used for imaging structures inside the body but has limited applications in the eye and superficial skin measurements.

High-frequency ultrasound (HFUS) uses frequencies from 20 to 100 MHz, increasing the spatial resolution of the images [298]. HFUS has been used in dermatology to characterise skin layer thickness and assess different skin conditions [299]. In ophthalmology, ultrasound biomicroscopy, a clinical HFUS implementation operating at 35–100 MHz, provides non-invasive, high-resolution imaging of the anterior segment, with a typical axial resolution of 20–50 μm [300,301].

Super-resolution ultrasound imaging (SRUS) employs an ultrafast programmable ultrasonic scanner and microbubbles as ultrasound contrast agents, typically ranging from 1 to 10 μm in diameter [302]. SRUS techniques, including ultrasound localisation microscopy, super-resolution fluctuation imaging, and structured illumination, allow distinguishing structures below the diffraction limit ($\sim\lambda/2$) that affect optical visualisation, see Section 3.1.1. These techniques have been widely used in microvasculature analysis [303].

Ultrasound Localisation Microscopy (ULM) has been tested using microbubbles (with radii 1–5 μm) in micro-channels and *in vivo* experiments on mice tumors. It achieved a maximum axial resolution of 6 μm and a lateral resolution of 40 μm, with simulations predicting resolutions of 5 μm axially and 11 μm laterally [218,304]. A theoretical model has been used to predict that in human breast imaging with a 7 MHz ultrasound probe, it is possible to achieve a resolution below 2 μm at an imaging depth of 30 mm, using sampling frequencies around 40 MHz [305]. ULM has been rated as a challenger to the traditional trade-off between resolution and penetration depth, highlighting its significant potential for clinical applications [306,307], by benchmarking microbubble localisation algorithms to identify those best suited for specific imaging scenarios.

Table 3
Typical measurement capabilities of the acoustic imaging method (AIM), spatial resolution (SR), interrogation area (IA), temporal resolution (TR) and frequency (f), see Fig. 9.

AIM	Sub-type	SR (μm)	IA (cm^2)	TR (ms)	f (MHz)
US	Conventional US	300–3000	1–100	1	2–10
	HFUS	30–50	0.1–10	10–100	20–100
	SRUS[B]	5 (axial) and 11 (lateral)[*]; 1[**]	0.01–1	0.1	10–40
AM	UBM	20–50	0.01–0.1	1×10^3	20–100
	SAM	15–20	0.01–1	1–10	<200
	CSAM	~5	0.001–0.1	1–10	<200
	AIM	~10	~0.04	1×10^4	1×10^3
	GHz-SAM[B]	~1	$(0.06–0.25) \times 10^{-2}$	1×10^4	2×10^3
	PU	~1 lateral and ~0.15 depth	<0.01	1×10^4	$10–1000 \times 10^3$
PAM	OR-PAM	0.2–10 (lateral); 15–30 (axial)	0.1–10	300	20–100
	AR-PAM	15–50 (lateral); 20–50 (axial)	0.1–10	30×10^3	1–20

[*] Microbubble ultrasound super-localisation imaging (MUSLI).
[**] Theoretical resolution limits [305] SR, IA and TR values correspond to typical literature demonstrations for each technique, unless otherwise indicated. Entries reflecting best-case performance or theoretical limits demonstrated in advanced prototype implementations are labelled "B".

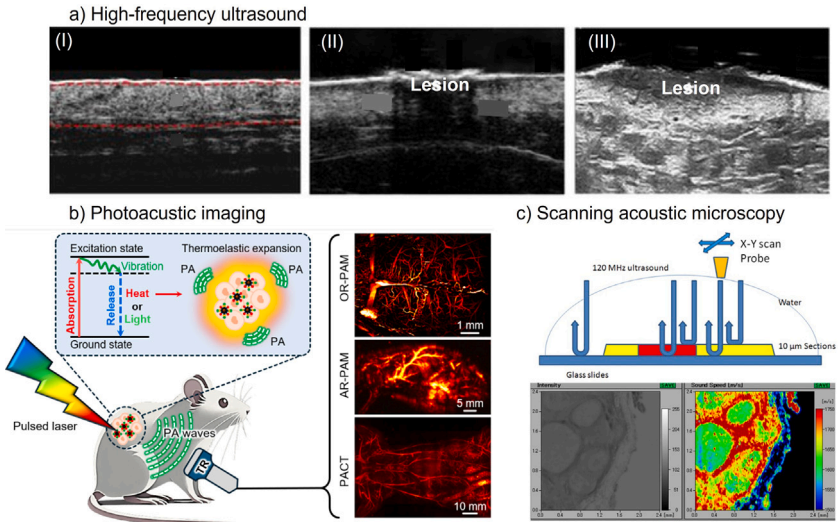


Fig. 7. Acoustic imaging methods for measuring mechanical properties. (a) HFUS imaging of skin layers showing (I) normal skin layers, (II) surface hyperkeratosis and (III) stratum corneum detachment [Image reprinted by permission from reference [308]. Copyright, 2021 John Wiley & Sons A/S]. HFUS provides high-resolution structural images (layer thickness, interfaces and echogenicity) that are often used as anatomical context for mechanical measurements and, when combined with active excitation (e.g., ultrasound elastography or acoustic-radiation-force methods), enable direct mapping of tissue stiffness by tracking tissue displacement or shear-wave speed. (b) Schematic of photoacoustic imaging modalities (OR-PAM, AR-PAM, and PACT) [Image reprinted from [333], which is licensed under CC BY 4.0]. Photoacoustics primarily reports optical absorption and vascular/functional contrast, but when combined with ultrasound detection or with co-registered ultrasound elastography it helps localise biomechanical heterogeneity (e.g., regions of altered perfusion or composition that correlate with stiffness). (c) SAM principles and example screenshot, showing measurement of acoustic intensity, speed of sound, attenuation, and thickness in tissue sections [Reprinted from [310], which is licensed under CC BY-NC-ND 3.0]. SAM measures acoustic velocity and attenuation at micron-to-millimetre scales; these acoustic parameters can be related quantitatively to elastic moduli and viscoelastic properties through mathematical modelling.

US limitations. Conventional US has limited spatial resolution, making it less suitable for characterising small-scale surface features. HFUS improves resolution but is inferior compared to optical methods. SRUS and ULM enhance resolution, but their reliance on contrast agents, complex instrumentation, and high computational demands can restrict their widespread application in clinical settings.

US in soft and biological materials. US techniques have been extensively applied to biomedical imaging, including assessing skin layer thickness and diagnosing dermatological conditions. Additionally, US is commonly used for imaging internal structures but has limited applications in the eye and in superficial skin measurements [299,308]. HFUS has been particularly useful in dermatology [299,308] (see Fig. 7(a)) and, as ultrasound biomicroscopy, provides non-invasive, high-resolution imaging of the anterior segment of the eye [300]. In ophthalmology, ultrasound biomicroscopy, a clinical HFUS implementation operating at 35–100 MHz, provides non-invasive, high-resolution imaging of the anterior segment, with typical axial resolution ≈20–50 µm [300,301]. SRUS has enabled detailed visualisation of microvasculature in organs, including the skin and the retina of the eye [303]. Microbubble ultrasound super-localization imaging (MUSLI) has potential for high-resolution imaging of tissue microstructures, with promising applications in oncology and vascular imaging [304].

4.1.2. Acoustic microscopy (AM) main features

Acoustic microscopy techniques enable high-resolution imaging and the study of material elastic and viscoelastic properties through the interaction of acoustic waves with a sample. Different modalities have been developed to enhance the resolution and contrast, providing valuable insights into biological and soft materials.

Conventional Scanning Acoustic Microscopy (SAM). In SAM, an acoustic beam is generated by a transducer and focused with an acoustic lens system, scanning the sample point-by-point. High-resolution images are generated by analysing the amplitude and phase of the reflected or transmitted sound waves [309]. A liquid immersion medium ensures good acoustic contact and a high refractive index for the acoustic lens [310] (see Fig. 7(b)). Unlike many optical methods, scanning acoustic microscopy does not rely on staining; image contrast instead arises from variations in acoustic impedance and the specimen's surface geometry [311,312].

Ultrasound Backscatter Microscopy (UBM). In UBM, the transducer transmits ultrasound pulses into the tissue and then receives the backscattered echoes. These signals are processed to create images reflecting the tissue's acoustic properties such as density and stiffness. Operating at 40–100 MHz typically yields axial and lateral resolutions on the order of 30–90 µm [313,314]. Its non-invasive nature makes it particularly suitable for ophthalmology, allowing tomographic imaging of ocular tissues [315].

Acoustic Impedance Microscopy (AIM). Acoustic impedance quantifies the resistance of a material to sound wave propagation, defined as $Z = v \cdot \rho$, where v is the sound speed and ρ is the density. In this technique, ultrasound waves travel through a thin plastic plate before interacting with the tissue, enabling impedance measurements [316]. This allows micro-scale imaging of tissues with high contrast and differentiation between normal and cancerous cells [317,318].

Scanning GHz Acoustic Microscopy (GHz-SAM). Higher acoustic frequencies improve resolution but their attenuation increases, requiring short focal length transducers. GHz-SAM achieves resolutions of about 1 µm using lenses with a 40 µm radius of curvature and water as a coupling fluid. However, manufacturing transducers for GHz-SAM is still problematic due to the difficulty of producing lenses with the necessary precision [319,320].

Limitations. SAM with high-frequency (> 300 MHz) requires precise coupling and liquid immersion, making *in vivo* applications challenging. UBM has limited penetration depth, typically less than 5 mm, restricting its use for deep tissue imaging. AIM requires careful calibration to ensure accurate signal interpretation. GHz-SAM requires advanced fabrication techniques for affordable high-quality transducers.

Applications in soft and biological materials. SAM has been used to study subcellular components, cell motility, elasticity, and tissue anisotropy [321]. It has provided quantitative measurements of bone and cartilage elasticity, benefiting biomechanical studies. In cancer research, SAM has enabled tissue classification based on stiffness differences [311].

Compared with optical methods, UBM can image deeper structures in the dermis and subcutis [322], which enables non-invasive, depth-resolved monitoring of topical and local administrations (e.g., microneedle patches, intradermal injections) by visualising formulation penetration, local oedema or tissue swelling, and structural changes in the dermis over time. Several commercial UBM systems are available and increasingly used in both research and clinical dermatology for lesion characterisation and preoperative assessment [323,324].

Acoustic impedance microscopy has been applied to micro-scale imaging of cerebellar tissue [317], 3D mapping of glial cells [325], and differentiation of cancerous cells [318]. More recently, it has been used for characterising tissue-mimicking materials for biomedical applications [326].

GHz-SAM is promising for non-destructive micro-elastic property assessment with micrometre resolution, useful in cellular mechanics and biomaterial characterisation [320]. In dermatological contexts, SAM has been used to examine neoplastic and inflammatory cutaneous tissue specimens, where ≈600–800 MHz SAM differentiated skin lesions on unstained sections [327]. It has also been applied to healing skin/wounds, correlating acoustic speed/attenuation with collagen and tensile strength of skin/wound tissue [328]. More recently, acoustic microscopy (>20 MHz) was combined with infrared spectroscopic mapping to enable early *in situ* melanoma detection in skin [329].

4.1.3. Photoacoustic imaging (PAI) main features

Photoacoustic imaging, or optoacoustic imaging, fuses optical contrast with ultrasonic detection to achieve high-resolution images at depths inaccessible to purely optical methods. A short non-ionising laser pulse is absorbed by tissue chromophores, producing a rapid thermoelastic expansion that launches an acoustic wave; this wave is then recorded by an ultrasonic transducer (or array) and reconstructed into an image [330,331]. Depending on the implementation, lateral spatial resolution can span from several hundred nanometres up to a few micrometres, while imaging depths typically reach the millimetre scale.

PAI is commonly grouped into photoacoustic tomography (PAT), photoacoustic microscopy (PAM), photoacoustic computed tomography (PACT), and photoacoustic endoscopy (PAE) [332,333] (see Fig. 7(c)). PAM itself is divided into two practical regimes:

Optical-resolution PAM (OR-PAM)—achieves lateral resolution on the order of hundreds of nanometres to a few micrometres but is limited to superficial imaging (typically < 1 mm) because optical focusing determines the resolution.

Acoustic-resolution PAM (AR-PAM)—achieves lateral resolution in the tens of micrometres while allowing substantially greater penetration (often an order of magnitude deeper than OR-PAM), as acoustic focusing determines the effective resolution [334].

PAI limitations. While PAI provides high sensitivity to chromophores such as hemoglobin and melanin [335], its effectiveness can be limited by factors such as the optical scattering properties of tissues, resolution constraints at deeper penetration depths, and the complexity of signal interpretation. Additionally, the requirement for precise laser and transducer alignment can pose challenges for *in vivo* applications [336].

PAI in soft and biological materials. PA imaging has found multiple applications in skin imaging, particularly in studies related to skin cancer [336], microvasculature [337,338], and burn injury assessment [339,340]. There is also a growing interest in using PAI to monitor drug delivery, assess penetration and efficacy [332,341–344], as well as in ophthalmology applications [345].

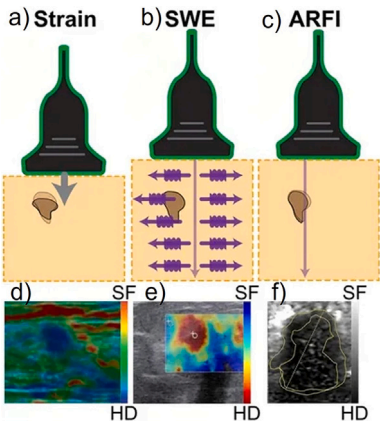


Fig. 8. Schematic representations (a–c) and corresponding ultrasound images (d–f) illustrating different elastography methods used to evaluate tissue stiffness. (a, d) Strain elastography relies on external compression to assess tissue deformation. (b, e) SWE measures the propagation speed of shear waves induced within the tissue. (c, f) ARFI imaging uses focused ultrasound pulses to generate localised tissue displacement [Image reprinted from reference [363]], licensed under CC BY 4.0.

4.2. Ultrasound Elastography (UE) main features

UE detects changes in the propagation of mechanical waves and links the response of tissues to external forces or vibrations with their elasticity [346]. The mechanical waves can be generated by an external mechanical input or by the ultrasound itself. UE can be divided into two categories, quasi-static and dynamic UE [347] (Fig. 8). In quasi-static UE, a constant stress is applied to the tissue, and the resulting strain is calculated using ultrasound image correlation. Using Hooke's law $F = -\kappa x$, with F as the applied force and x as the displacement caused by F, this technique offers an indirect assessment of tissue stiffness κ [348]. However, the lack of knowledge about the applied stress prevents the accurate quantification of the local Young's modulus [347]. Dynamic UE applies a time-varying force to excite shear waves, whose speed c_s depends on the shear modulus G and tissue density ρ,

$$c_s = \sqrt{\frac{G}{\rho}}, \tag{1}$$

which relates to the Young's modulus by the Poisson ratio, see Section 1. By measuring c_s, it is possible to predict E values in a non-invasive manner [347]. Below we outline various techniques used in UE; further details about the propagation of ultrasound waves in soft solid media are found elsewhere [349–351].

Acoustic Radiation Force Impulse Imaging (ARFI) and Shear Wave Elasticity Imaging (SWEI) are two of the most commonly used methods in ultrasound elastography. ARFI employs the acoustic radiation force to displace tissue, followed by ultrasound imaging to detect and track the resulting displacements [352,353]. Although ARFI provides information on the elasticity and viscosity of tissue at the focal point, it lacks the capacity to provide a complete quantitative estimate of Young's modulus. SWEI, in contrast, generates shear waves within tissues using ARF and ultrasound imaging to track the propagation of these waves [354,355]. The displacement map obtained indicates the stiffness distribution within the target soft tissue, which is assumed to be directly proportional to the elastic modulus of the local tissue [346]. This system has the advantage that the deformation and wave detection are both provided by an ultrasound probe, eliminating the need for an external source of vibration [346].

Transient Elastography (TE) generates low-frequency (≈50–500 Hz) transient waves in tissue using an external vibrator and tracks their propagation to infer mechanical properties [346,356–358]. Under the

assumption that the sample behaves elastically and that its size is large compared to the induced wavelength, TE yields quantitative estimates of tissue elasticity [346]. Supersonic Shear Imaging (SSI) builds on these principles but employs focused ultrasound beams to create a moving ARF, resulting in interfered shear wavefronts [359]. Ultrafast imaging tracks the propagation of these waves, allowing for the measurement of shear wave velocities and the determination of elastic moduli in soft tissues [360].

Guided Wave Elastography (GWE) is specifically designed for thin-walled soft tissues, where waves are guided and strongly dispersive due to wall thickness comparable to the wavelength [346]. Focused ARF generates guided waves in these tissues, whose dispersion relation

$$\omega^2 = \frac{G}{\rho} k^2 \tanh(kh), \tag{2}$$

relates angular frequency ω, wavenumber k, and material properties G, ρ and h tissue thickness, which can be inferred by fitting measured $\omega(k)$ curves [346,361,362].

UE limitations. The accuracy of ARFI and SWEI is influenced by the inhomogeneity of biological tissues, as variations in local density and mechanical properties can introduce artifacts in elasticity maps [346]. The estimation of Young's modulus using SWEI relies on the assumption of isotropy, which may not hold true for many biological tissues that exhibit anisotropic behaviour due to their fibrous heterogeneous structure [347]. Additionally, transient elastography assumes that tissues behave as purely elastic materials, which neglects their viscoelastic nature and may lead to inaccuracies in stiffness estimation [360].

SSI requires ultrafast imaging capabilities, which may not be available in standard ultrasound systems, limiting its accessibility for clinical applications [364]. GVE faces challenges related to the complexity of wave dispersion in thin tissues, requiring sophisticated numerical and theoretical models to extract accurate mechanical parameters [361,362,365].

UE in soft and biological materials. Ultrasound elastography has been widely used in clinical diagnostics and tissue characterisation. ARFI has been applied to measure liver fibrosis, where changes in tissue stiffness serve as indicators of disease progression [352,353]. SWEI has been successfully employed to assess muscle elasticity and stiffness changes in neuromuscular disorders [354,355]. Transient elastography has been extensively used in breast cancer detection, as malignant tumors often exhibit increased stiffness compared to healthy tissue [358].

SSI has demonstrated potential for monitoring brain elasticity in neurological diseases, such as hydrocephalus, where changes in intracranial pressure can alter brain stiffness [360]. Guided Wave Elastography has been explored for measuring the mechanical properties of arteries, providing insights into vascular diseases and early-stage atherosclerosis detection [361,362,365].

Overall, ultrasound elastography offers a non-invasive, versatile approach not only for disease diagnostics and tissue engineering, but also for drug-delivery applications. For example, it can be used to assess barrier integrity and local stiffness that influence transdermal uptake, to guide and monitor the dispersion of locally injected formulations, and to report early mechanical changes that indicate tissue response to therapy.

4.3. Summary of acoustic imaging measurement techniques

Table 3 and Fig. 9 provide an overview of the acoustic imaging modalities discussed in this review. Table 3 compiles representative spatial resolution (SR), interrogation area (IA), temporal resolution (TR), and operating frequency ranges for US, AM and PAM, together with their principal subtypes. Fig. 9 summarises the practical operating characteristics of selected acoustic modalities and the principal trade-offs relevant for dynamic mechanical measurements in soft tissues. SAM and

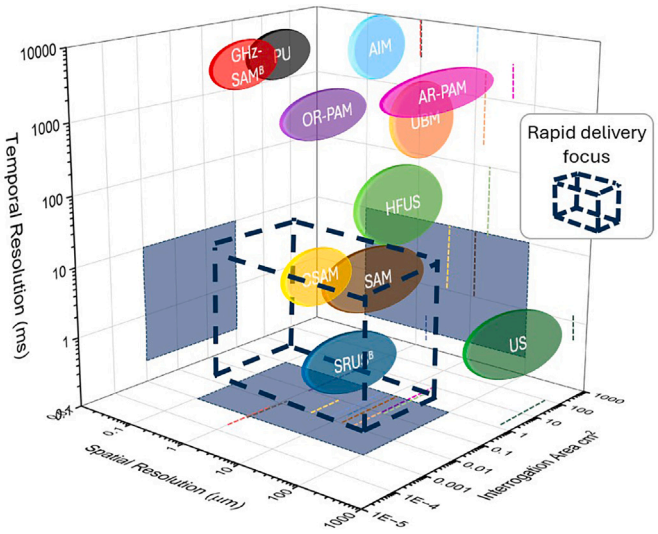


Fig. 9. Practical operating characteristics of acoustic imaging modalities (spatial resolution, temporal resolution, interrogation area/depth). Coloured ellipsoids indicate typical, representative operating regions (not instrument-specific limits). SAM/GHz-SAM offers micron to sub-micron lateral resolution at high acoustic frequencies but has limited penetration and field of view and relatively slow point-scanning acquisition modes [367,372]. HFUS (tens of MHz) gives improved spatial resolution (tens to a few hundreds of μm) with millimetre penetration and rapid frame rates suited to millisecond dynamics [368]. ULM/SRUS attains sub-diffraction resolution (<10 μm) by localising microbubble contrast, at the cost of requiring contrast agents, ultrafast imaging and aggregation over many frames [303,366]. OR-PAM/AR-PAM spans optically limited high lateral resolution (on the order of a few micrometres) to acoustically limited deeper imaging (coarser resolution), making PAM useful when optical absorption contrast is required [369,370]. Conventional clinical US covers large volumes with lower spatial resolution but very high frame rates, favouring bulk and rapid dynamic elastography over subcellular detail [371]. SR, IA and TR ranges correspond to typical literature demonstrations for each technique, unless otherwise indicated. Entries reflecting best-case performance or theoretical limits demonstrated in advanced prototype implementations are labelled "B"; further details are given in Table 3.

GHz-SAM use focused, high-frequency acoustic beams and can reach micron to sub-micron lateral resolution at frequencies up to the GHz range; however, these high frequencies substantially limit penetration depth and the practical lateral scan area, and conventional SAM implementations rely on mechanical point-scanning which constrains volumetric acquisition speed [366,367]. HFUS (tens of MHz) provides a compromise of improved spatial resolution (tens to a few hundreds of micrometres) with millimetre penetration and substantially faster frame rates compatible with millisecond-scale dynamic studies in dermatology and soft tissue imaging [368]. SRUS/ULM overcome the diffraction limit by localising microbubble contrast agents; they can resolve microvascular structures below ≈10 μm in favourable conditions, but require ultrafast imaging, contrast administration and accumulation over many frames (often seconds to minutes of acquisition), which limits real-time capability for some applications [303,366]. PAM splits into OR-PAM —optically limited lateral resolution down to sub-micrometre–micrometre scales with shallow (~1 mm) penetration—and AR-PAM, which reaches deeper but with coarser resolution; PAM therefore occupies an intermediate region of the resolution–depth–speed space and is particularly attractive when optical absorption contrast is required [369,370]. In contrast, conventional clinical US covers large interrogation volumes with lower spatial resolution (sub-millimetre to millimetre) but very high frame rates (tens to thousands of fps depending on mode), making it suitable for fast dynamic elastography and bulk motion measurements at the expense of fine spatial detail [371].

5. Contacting solid methods

We focus here on mechanical testing techniques that are applicable on the microscale, highlighting the more traditional mechanical tests, relying on the skin or the eye being contacted by a solid. In these methods the internal system read-out requires no additional data processing or visualisation techniques.

5.1. Indentation probes (IP) main features

Indentation is a widely used technique for evaluating the mechanical properties of materials. It involves pressing a probe—such as a sharp or spherical indenter—into the surface of a material and measuring the resulting deformation. By monitoring the indentation depth as a function of the applied load, various mechanical properties can be extracted, including Young's modulus, hardness, and yield strength (see Introduction). This technique is, in principle, non-destructive and can be applied *in vivo*. A wide variety of probes are available, made from different materials and spanning characteristic dimensions from μm to nm. These probes can achieve force resolutions as low as one-billionth of a Newton and depth resolutions ranging from 1 nm to several cm. Although numerous reviews [373–377] and books [378,379] have extensively described this technique, we focus here on its key characteristics and its application for measuring properties in selected hydrogels, the eye, and the skin.

The technique has a long history, with seminal works based on Hertz contact law and Sneddon's equation for analysing the force-displacement curves [380–383]. Knowledge of the contact area is crucial for correctly correlating the unloading curves with material properties. The contact area depends on the probe shape and the material characteristics. For example, when a conical indenter probes a flat solid material perpendicularly, the relationship between the load (F) and penetration depth (h) follows a square relationship [384]:

$$F = \frac{2Eh^2}{(1-\nu^2)\pi}\tan(\theta) = k_e h^2 \tag{3}$$

where k_e is a proportionality factor that depends on the elastic properties of the measured material $E/(1-\nu^2)$ and the indenter geometry θ. The square relationship between load and penetration depth applies to homogeneous solids that lack a hierarchical structure.

5.1.1. IP limitations

If the material is not homogeneous, the loading behaviour deviates from the square relationship due to size effects and intrinsic length scales such as cells, hairs, and different material layers. The variations in measured hardness of biomaterials depend on the diameter of the probe [384]. Size effects are also observed in metals, ceramics, polymers, and biological tissues [354]. Common causes include surface films, surface roughness, indenter rounding, and loading conditions [385]. For soft solids, detecting the contact point and avoiding sample piercing are additional challenges [386]. Moreover, surface energy can influence indentation at small scales, with elastocapillary forces becoming significant [75]. Unlike metals, where surface tension effects occur at sub-nanometre scales, soft solids exhibit these effects at micron scales [387].

Variations in elastic modulus and hardness depend on probe size, indentation depth, speed, and applied force. Generally, increasing indentation speed and force results in higher elastic modulus values, while increasing probe size and depth lowers the measured elastic modulus and hardness. These dependencies are illustrated in Fig. 10.

5.1.2. IP in soft and biological materials

The versatility of indentation allows IP to determine residual stresses, anisotropic properties, and property gradients. Recent advancements have integrated indentation with multifidelity deep learning, improving property extraction accuracy [388].

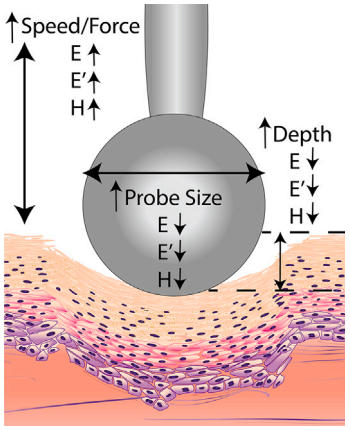


Fig. 10. Influence of loading speed, applied force, probe size and indentation depth on the measured mechanical properties of materials. Increasing the loading speed and applied force generally results in higher measured elastic modulus and hardness, whereas increasing the probe size and indentation depth tends to reduce the measured elastic modulus and hardness.

For microscale skin indentation, probe sizes between 0.5–100 μm have been used at depths of 0.5–80 μm with speeds of 1–100 μm/s and forces of 1–2 N. These conditions yield elastic moduli values between 2 kPa and 30 MPa [389–391]. Other parameters of importance include reduced elastic modulus and hardness, with lower variation ranges. For a summary of the values reported under different conditions, see Table 4 and Fig. 10. The effect of probing force is often not explicitly detailed in the cited studies, although it plays a crucial role: for example, it can influence the probing speed when depth is held constant, or affect the indentation depth when speed is kept constant.

5.2. Atomic Force Microscopy (AFM) indentation main features

Atomic Force Microscopy systems are widely used for surface topography measurements [392]. When AFM tips are modified with microspheres, they can function as micro-indentation systems. Commercially available micro-sphere AFM tips facilitate these measurements, providing insights into the mechanical properties of biological tissues at the microscale.

The measurement principles of AFM-based indentation are similar to traditional indentation techniques, but they involve smaller probe sizes (ranging from 1.9 to 20 μm) and detect forces in the range of 30 to 400 nN. The indentation depth typically reached is between 200 and 500 nm, and probes only stratum corneum (SC) deformations [393].

5.2.1. AFM limitations

While AFM indentation provides high-resolution mechanical property measurements, its application is constrained by several factors. The limited indentation depth restricts its ability to probe deeper layers of the skin, such as the dermis. Additionally, experimental parameters such as the spring constant of the AFM tip, the adhesive properties of both the sample and the modified probe material, and potential artefacts may influence measurement accuracy. Moreover, while unmodified AFM tips with apex radii as small as 10 nm can probe tissues at an even smaller scale [393], these measurements fall outside the scope of this review, which focuses on techniques capable of resolving mechanical properties at mesoscopic scales (micrometre to millimetre range) and relevant timescales for dynamic biological processes. For readers interested in nanoscale mechanical probing using AFM, we refer to comprehensive reviews [394,395].

5.2.2. AFM in soft and biological materials

AFM-based indentation has been used to characterise the mechanical properties of the skin layers. One study reported the elastic modulus of

Table 4
Techniques used to determine selected parameters (Elastic modulus (EM), Reduced Elastic modulus (REM) and Hardness (H)) and their reported value ranges for the mentioned experimental parameters (probe size, probe force, probing depth and probing speed).

Technique	Tissue	Parameters	Probe radius (μm)	Force (μN)	Depth (μm)	Speed	Measured value (MPa)
AFM	cornea*	EM	1–68	$0.9–20 \times 10^{-3}$	0.1–20	2–15 μm/s	$6.1–800 \times 10^{-3}$
	skin**	EM	2–6	30–400	0.2–0.5	1–100 μm/s	7×10^{-3}–13
Microindentation	cornea*	Effective EM	500	20–50	<60	0.67–1.67 μm/s	$19–52 \times 10^{-3}$
	skin**	EM	0.5–100	$1–2 \times 10^{6}$	15–80	1–100 μm/s	2×10^{-3}–30
		REM	1–20	100–500	0.5–1	20 μN/s	$0.21–0.87 \times 10^{3}$
		H	1–20	100–500	0.5–1	20 μN/s	0.5–14.5

* anterior corneal stroma.
** stratum corneum and epidermis.

the SC as 7 MPa to 13 MPa [396], while another investigated the dermis, reporting values between 1GPa to 10 GPa [397]. Although direct comparisons with traditional indentation techniques are scarce, the reported values follow similar trends (see Table 4). Similarly, AFM-based indentation approaches have been applied in ophthalmology to characterise the micromechanical properties of ocular tissues such as the cornea, sclera, and lens capsule, providing insights into age-related stiffening, disease progression (e.g., keratoconus, glaucoma), and the effects of cross-linking treatments [398–400].

5.3. Dynamic indentation (DI) main features

Dynamic indentation methods use a varying load, unlike static indentation where the load remains constant for the duration of the test. In this technique, an oscillatory displacement, such as a sinusoidal temporal wave, is applied to the probe, allowing the assessment of viscoelastic properties, including storage and loss moduli, under dynamic conditions [401]. These moduli are determined by analysing the material's in-phase and out-of-phase responses to dynamic loading.

The frequency at which indentation is performed influences the measured viscoelastic properties of soft materials [402]. At higher frequencies, materials tend to appear stiffer due to limited time for molecular rearrangement, while lower frequencies allow for more relaxation and thus softer responses. In dynamic indentation methods, an oscillatory load is applied, generating elastic waves whose propagation is analysed to extract mechanical parameters such as storage and loss moduli [403]. The resulting displacement field—i.e., the movement of material points in response to the wave—can be tracked using appropriate sensors. Quartz transducers are often employed to measure dynamic loads due to their piezoelectric nature, which generates an electrical signal proportional to the applied force.

5.3.1. DI limitations

One of the main challenges of dynamic indentation is to ensure a consistent and precise contact between the probe and the sample, especially for soft and biological materials that may deform significantly under loading. Additionally, variations in hydration, temperature, and surface roughness can influence the mechanical response thus, proper calibration is essential. The frequency range of oscillation must be carefully chosen to avoid artifacts due to inertial effects or resonance phenomena, particularly when measuring thin tissues or layered structures. Furthermore, while dynamic indentation provides valuable insights into viscoelastic behaviour, it often requires complementary techniques, such as imaging or other rheological methods, to fully characterise complex biological materials.

5.3.2. DI in soft and biological materials

Dynamic indentation has been widely applied in the study of skin mechanics. *In vivo* studies on subjects aged 18 to 70 have employed strain amplitudes ranging from 10 to 60 Hz and displacements from 1 to 10 μm using piezoelectric devices, showing that skin becomes less elastic and more viscous with age [404]. Mechanical stimulations at frequencies in the 0.5–30 kHz range have been applied to the skin and the response measured with piezoelectric transducers for assessing anisotropy and age-related changes in skin stiffness and damping behaviour [405]. Another study used an annular vibrator to measure storage and loss moduli at frequencies between 200 and 1000 Hz [406] (see Fig. 11(a)). Surface waves on tissue have been generated using a mechanical shaker with a force of 1 N at frequencies between 100 and 400 Hz, with propagation measured with laser vibrometry [407]. This system has been proposed for assessing tissue health based on viscoelastic properties, incorporating ultrasound transducers to measure wave propagation in deeper tissue layers. Similarly, a device equipped with a voice coil can excite the skin surface at specific frequencies, tracking the displacement with an accelerometer [408].

These dynamic measurements have direct relevance for drug-delivery applications. The frequency- and depth-sensitivity of indentation and surface-wave methods help to characterise barrier integrity (for example, stratum corneum stiffness and hydration) and to detect rapid, formulation-induced mechanical responses that correlate with enhanced permeation or skin uptake.

In contrast, for the eye, dynamic indentation with solid probes has been scarcely explored; possibly because it carries a non-negligible risk of being invasive.

5.4. Ballistometry tests (BT) main features

Ballistometry tests work by impacting a sample with a solid projectile, usually a sphere, and then measuring the deformation and rebound characteristics of the sample upon impact [411]. By analysing the interaction between the projectile and the material, various mechanical properties such as stiffness, damping, and viscoelastic behaviour can be determined [409,412] (see Fig. 11(b)). This technique provides insights into the dynamic response of materials under impact, making it particularly useful for understanding how soft materials behave under sudden loading conditions (typically milliseconds), mimicking real-world scenarios where tissues or gels experience rapid changes in force or impact [411].

While traditional ballistometry employs centimetre-scale probes and primarily measures the restitution coefficient, advancements in this technique could involve adapting it to micrometre-scale probes for higher precision deformation measurements. Additionally, integrating ballistometry with visualisation techniques, such as those outlined in Section 3, would allow for real-time tracking of both the restitution coefficient and the dynamic deformation of the skin upon impact.

5.4.1. BT limitations

A primary limitation of ballistometry is its sensitivity to external factors such as humidity and temperature, which can affect the restitution coefficient [413]. Additionally, most implementations employ centimetre-scale probes, which may not capture small-scale heterogeneities in biological tissues, on the order of micrometres. The dynamic nature of the BT requires HS imaging (see Section 3.1.1) or precise

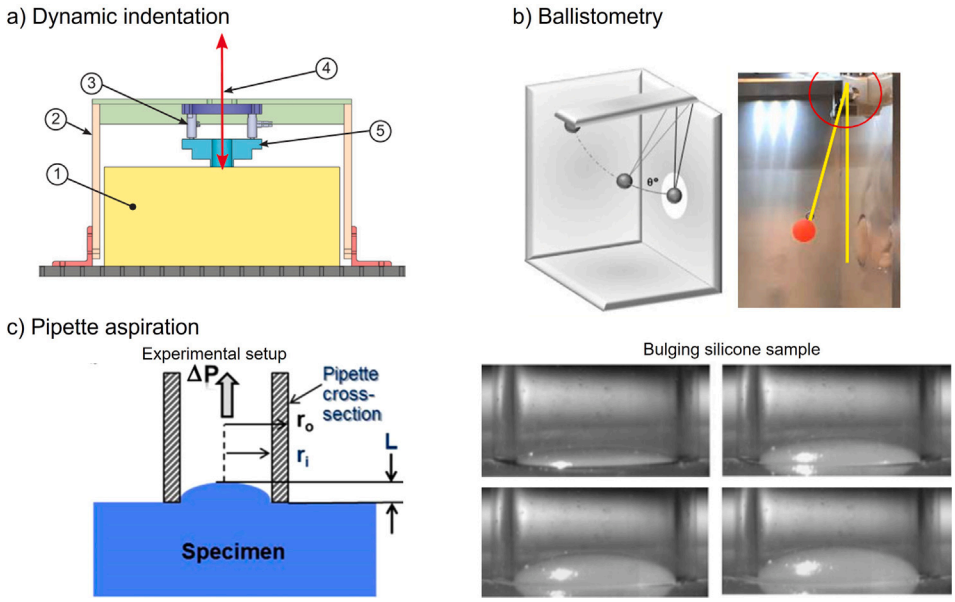


Fig. 11. (a) Dynamic indentation. A schematic of the experimental setup, including the sample location (1), frame (2), preloaded piezoelectrics (3), SLDV beam (4), and annular mechanical radiator (5), used to measure material response to indentation [Reprinted by permission from reference [406]. Copyright 2015, IOP Publishing]. (b) Ballistometry. A device measures skin elasticity based on the elastic angle. A rubber ball of ≈5 cm in diameter is dropped, collides with skin or other biological material, and the elastic angle (the maximum rebound angle) is measured [Image adapted from reference [409], which is under CC BY-NC-ND 4.0]. (c) Pipette aspiration testing. Images show increasing aspirated length of a silicone sample with increasing aspiration pressure. The material bubble is centred within the pipette lumen [Reprinted by permission from reference [410]. Copyright 2015 Elsevier Ltd.].

sensors to capture rapid deformation events, adding complexity to the setup.

5.4.2. BT in soft and biological materials

Ballistometry has been used since the 1970s to assess the elastic properties of the skin *in vivo*, with studies covering human subjects ranging from 8 to 80 years old [413]. These investigations demonstrated that the coefficient of restitution (COR, defined as the ratio of rebound velocity to impact velocity) decreases with age [413]. COR values were around 0.75–0.80 in young adults versus 0.40–0.50 in elderly subjects [414]. Subsequent research has expanded its use in characterising skin properties [409,415], confirming that skin stiffness is the predominant measurable parameter, which ballistometric studies have reported in the range of approximately 0.5–2.0 kPa depending on anatomical site and subject age [416]. Furthermore, relative humidity levels have been shown to not influence the coefficient of restitution of facial skin [413]. In addition to human skin, ballistometry has been used to assess the suitability of porcine skin as a surrogate for human skin in experimental settings [417].

Ballistometry traditionally uses centimetre-scale probes to measure only the restitution coefficient. By switching to micrometre-scale probes, the same method could precisely capture much smaller deformations. Furthermore, combining ballistometry with one or more visualisation techniques from Section 3 would allow simultaneous measurement of the restitution coefficient and real-time skin. In principle, combining ballistometry with real-time imaging could also be adapted to study ocular tissues, although such applications would require major modifications to comply with current safety and comfort standards and thus remain at a conceptual level.

5.5. Suction-based systems (SBS) main features

Pipette aspiration, originally developed for cellular membrane analysis, provides a method to evaluate the mechanical properties of tissues and cells [47,418]. In this technique, a material is partially drawn into a micro-sized glass pipette using controlled negative pressure (usually 0.1–10^5 Pa) [410,419] (see Fig. 11(c)). The resulting deformation is tracked, enabling the determination of material properties such as elastic modulus and viscosity. The cutometer, another suction-based device, operates on a similar principle but at macroscopic length scales. It applies controlled negative pressure to a defined skin area through a circular probe and records deformation over time in real-time, typically within seconds.

5.5.1. SBS limitations

Pipette aspiration offers advantages such as non-invasiveness, ease of use, time efficiency, and high sensitivity. The method is generally reversible; however, excessive pressure can cause permanent deformation in materials exhibiting plastic behaviour. As a contact-based method, there is also a risk of cross-contamination. In contrast, while the cutometer is widely used for skin analysis, its applicability is limited to macroscopic measurements, and its sensitivity to small-scale heterogeneities in the skin may be lower compared to microscale techniques.

5.5.2. SBS in soft and biological materials

Pipette aspiration has been used to determine the mechanical properties of cell membranes but has also been applied to measure the nonlinear mechanical behaviour of arteries, multilayered materials, and solid tissues [410,420–422]. The technique has been particularly useful in studying tension-induced morphogenesis, providing an alternative to more complex and expensive methods like AFM or optical tweezers. The cutometer, on the other hand, has been extensively used in dermatological studies to assess changes in skin properties due to ageing, pathological conditions, or external treatments such as topical creams, moisturisers or UV exposure [423–426]. Consequently, cutometer measurements are useful functional endpoints in topical/transdermal drug-delivery work—helping to quantify formulation-induced mechanical changes, monitor barrier disruption and recovery, and inform dose or formulation selection *in vivo*.

5.6. Tensile and compressive tests (TCT) main features

Tensile and compressive tests are common methods for characterising the mechanical behaviour of materials under different types of loading. Similar to indentation methods, these tests involve applying forces to a material specimen and measuring its response to the applied load [427].

In a tensile test, a specimen is pulled apart along the deformation axis, applying a tensile force that stretches the material. Conversely, in a compressive test, a specimen is subjected to an axial force that compresses the material. These tests determine various mechanical properties, such as Young's modulus, yield strength, ultimate tensile strength, compressive strength, and strain. Tensile testing can also be performed biaxially, where a specimen is stretched along two perpendicular axes, providing insights into the material's anisotropic behaviour [428].

A widely used dynamic testing apparatus is the split Hopkinson pressure bar (SHPB), designed for high strain rate and impact loading studies [429]. The SHPB system consists of an incident bar, a striker bar, and a transmitted bar, typically made of high-strength materials to withstand the high stresses and strain rates during the experiment. The stress-strain behaviour of the material under dynamic conditions is determined by analysing incident and transmitted stress waves [430].

5.6.1. TCT limitations

Tensile and compressive tests primarily characterise the bulk response of the material and do not provide information about its hierarchical structure [427]. Additionally, soft materials present challenges for SHPB testing due to their low stiffness and high deformability, which can lead to premature failure and inaccuracies in stress-strain measurements [430]. Achieving dynamic stress equilibrium is necessary for obtaining valid results, and inertia effects from the high-momentum striker bar can further complicate measurements [431].

Tensile and compressive testing methods were initially developed for microelectromechanical systems (MEMS) [432,433]. The fundamental challenge in miniaturising these conventional techniques lies in delivering and accurately measuring the forces and displacements without introducing artefacts. At the microscale, alignment tolerances become critical: even slight misalignment between grips can generate bending moments or uneven stress fields. Likewise, parasitic forces—such as capillary, electrostatic, or friction forces at the grips—can overwhelm the signal of interest. Precise calibration of force transducers and displacement sensors is also difficult, as thermal drift and sensor compliance can introduce significant errors. Despite these difficulties, microgrippers with nN resolution have been developed to compress microscale PDMS samples [434].

5.6.2. TCT in soft and biological materials

Microscale tensile tests have been used to characterise the mechanical properties of *ex vivo* skin [435]. These tests reveal that the ultimate tensile strength of skin is strongly influenced by crack pathways, leading to significant variability even among samples from the same source. The anisotropy and inhomogeneity of skin tissue play an important role in these variations.

In addition, a piezoelectric-based device has been proposed for localised tensile testing of skin [436]. This handheld system applies stress tangentially using piezoelectric actuators, enabling measurements of elasticity, viscosity, surface stiffness, relaxation, creep, and hysteresis in soft materials. The device achieves surface strains ranging from 0.02 to 0.16 under forces of 0.1 to 0.75 N, and has been shown to be effective in characterising soft tissues with elastic moduli in the range of tens to hundreds of kilopascals —typical for human skin and other biological tissues. This makes it a promising tool for *in vivo* mechanical testing across physiologically relevant ranges and for drug-delivery research, since localised measurements of barrier stiffness and viscoelastic response can guide predict uptake, and monitor treatment-induced changes in tissue mechanics.

5.7. Shear rheology (SR) main features

Shear rheology is a powerful technique for measuring the mechanical properties of soft materials under shear deformation. This technique is particularly interesting for probing the viscoelastic response of a material and can provide its shear modulus, viscosity, and viscoelastic relaxation times [437].

Shear rheology measurements are commonly performed with a rotational rheometer, which typically consists of a parallel plate or concentric cylinder geometry. The soft material is placed between the plates or cylinders, and a controlled shear stress or shear rate is applied. The resulting deformation is monitored, and the shear stress and strain are measured as a function of time or strain. By analysing the stress-strain relationship, the shear modulus can be calculated [438].

The simplest rheometer has two parallel plates as the basic geometry. Within the plates, the strain is not uniform, starting at zero on the rotation axis and increasing linearly toward the edge of the parallel plates. Despite this non-uniformity, if the material exhibits a linear viscoelastic response, the stress–strain relationship can still be reliably extracted. This is possible because stress remains proportional to strain, allowing the use of the measured torque to compute stress via the integral $\tau = \int_0^R \sigma(r) 2\pi r^2 , dr$, where $\sigma(r)$ is the local stress at a radial distance r, and R is the radius of the plate. This approach is applicable to simple, compositionally homogeneous materials, where the local mechanical response does not vary across the sample. Additionally, the global stress–strain behaviour obtained from this setup is analogous to that of uniaxial tensile or compression tests, making it a useful comparative method for evaluating viscoelastic parameters. Alternative geometries, such as cone-and-plate and Couette configurations, maintain constant shear stress throughout the sample and are often preferred when a uniform stress distribution is required [438].

Rheometers can be stress-controlled and strain-controlled. In a stress-controlled rotational rheometer the instrument imposes a prescribed, time-varying torque on the measuring geometry and records the resulting angular displacement (and derived shear strain) of the sample as a function of time [77,439]. The angular displacement (and rotation rate) is typically measured with high-resolution encoders or non-contact optical displacement systems, which permit precise strain tracking without mechanical interference with the sample [77]. Contact-free optical detection is commonly implemented via laser interferometry; for example by monitoring laser light reflected from a micropatterned disc attached to the rotating tool. This technique enables sub-micrometre resolution of angular motion [77]. In contrast, in a strain-controlled rheometer the motor prescribes the angular displacement (or rotation rate) of one tool over time while a calibrated torque transducer at the opposing tool measures the sample's resulting torque (stress), allowing direct determination of stress under a controlled deformation protocol [77,438].

5.7.1. SR limitations

Shear rheometry in soft materials relies on assumptions such as homogeneous stress distribution and ideal boundary conditions, which often fail for highly structured or heterogeneous materials. Additionally, wall slip and liquid sample evaporation can affect measurement accuracy, especially for low-viscosity or highly deformable samples like the eye and skin [440].

Another challenge arises when working with very small sample volumes, where conventional rheometers may not provide sufficient sensitivity.

5.7.2. SR in soft and biological materials

A particularly useful test for characterising soft materials is to apply a sinusoidal shear strain $\epsilon_{xy} = \epsilon_0 \sin(\omega t)$. Under these conditions, the stress is assumed to vary sinusoidally as well, i.e., $\sigma(t) = \sigma_0 \sin(\omega t + \phi)$. The component of stress that is in phase with ω corresponds to the elastic response, while the out-of-phase component (viscous) varies like the

strain rate $\dot{\epsilon}_{xy}$. From this analysis, the complex modulus can be defined as:

$$G^* = G' + iG'', \tag{4}$$

where $G' = \sigma_0 \cos\phi$ and $G'' = \sigma_0 \sin\phi$. The loss angle, defined as $\tan\phi = G''/G'$, provides a measure of the ratio of energy lost to energy stored during cyclic deformation [440].

To address this, microrheology techniques have been developed, offering an alternative approach for microscale measurements [441]. Microrheology has been employed to measure the mechanical properties of *ex vivo* skin explants [441]. In this method, the skin explants are placed on a piezoelectric actuator that induces lateral motion, applying shear stress to the sample. By analysing the response of the sample to the oscillatory movement of the piezoelectric actuator, the shear complex modulus can be determined as a function of the oscillation frequency [441]. Microrheology is particularly relevant at the microscale because it probes length scales comparable to the thickness of individual skin layers (e.g., the stratum corneum and epidermis), capturing local heterogeneities that bulk rheometry cannot resolve. Additionally, the small sample volumes and high-frequency range accessible with piezoelectric actuation make microrheology well suited for studying dynamic processes—such as rapid deformation or local fluctuations—in soft biological tissues.

6. Contacting fluid methods

Contacting fluid methods apply a controlled load to the tissue using a liquid or a gas—either as a pressure difference (bulge/suction) or as a jet/flow that produces localised indentation and shear. The load is prescribed via the pressure difference, flow rate, and nozzle/port geometry, yielding either distributed or highly localised deformations at mesoscopic scales. In practice, these tests are commonly coupled to optical readouts to record the spatiotemporal displacement field. Compared with solid-contact probes, fluid loading is non-contact or soft-contact, reducing surface damage and probe–sample artefacts. The spatial resolution is set primarily by the aperture/nozzle (typically tens to hundreds of micrometres), and the temporal regime can range from quasi-static to high-rate transients. Reliable and comparable results require careful calibration of pressure/flow and boundary conditions, together with an appropriate inversion model. The following sections describe bulge testing and flow elastography as representative implementations.

6.1. Bulge testing (BGT) main features

BGT is a technique used to determine the mechanical properties of thin films and membranes by applying a biaxial stress field. The material of interest is clamped at its edges, creating a confined area that expands when subjected to internal pressure [442] (see Fig. 12(a)). After controlled pressure is introduced into the liquid channel, the deformation of the sample is monitored and the strain is measured. This method is particularly effective for evaluating elastic properties in thin metallic and polymer films modelled as membranes, where radial stresses are negligible [443–445].

The stress calculation assumes that the deformed membrane adopts a spherical cap shape, with principal stretches determined by differences in arc lengths or gradients of displacements [446]. Stress is then calculated from the applied pressure, considering the deformed shape and ignoring bending effects [447]. For materials whose thickness is not negligible compared to the radius of curvature and that undergo large deformations, an inverse finite-element approach has been used to compute properties from pressure-displacement bending [448].

6.1.1. BGT limitations

BGT has limitations when applied to materials thicker than 200 μm, as membrane-based calculations typically disregard bending effects. This assumption holds for thin membranes in the range of 10 μm to 100 μm. However, for samples exceeding the 100 μm in thickness, bending becomes non-negligible, and simplified membrane models

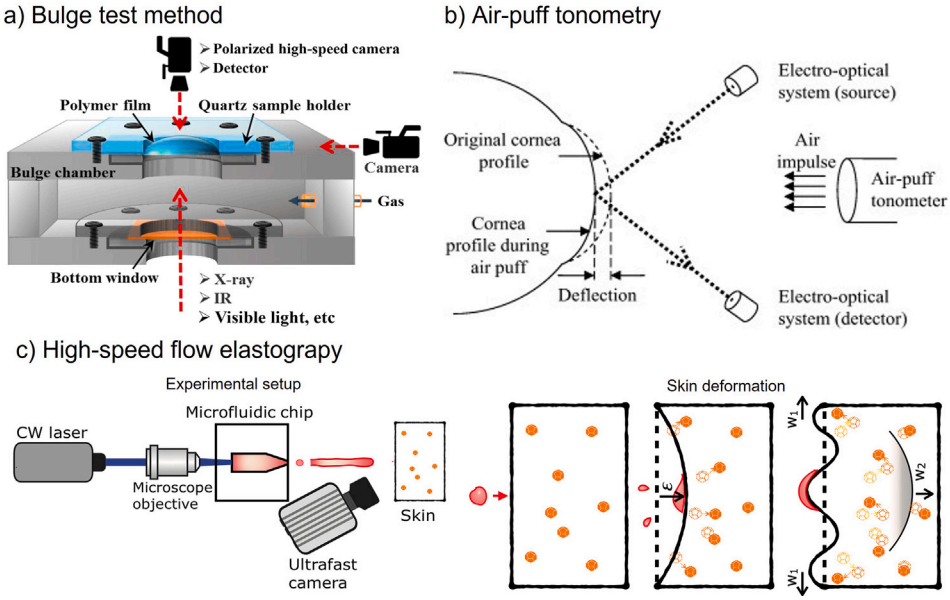


Fig. 12. Experimental techniques for characterizing the mechanical response of soft materials using fluids as the probe. (a) Schematic of the bulge testing method used to determine mechanical properties of thin films by pressurizing a polymer film clamped over a chamber and capturing deformation with optical or X-ray imaging [Reprinted by permission from reference [442]. Copyright 2019 Elsevier Ltd.]. (b) Schematic diagram of an air-puff tonometer. A brief, calibrated air pulse induces transient corneal deformation; an optical detector (or pressure transducer) measures applanation/rebound timing and amplitude, from which intraocular pressure is inferred [Reprinted by permission from reference [455]. Copyright 2011 IPEM]. (c) Experimental workflow for high-speed elastography via microfluidic jet impact driven by thermocavitation. A continuous-wave laser focused through a microscope objective creates a vapour bubble in a microfluidic chip, launching a jet toward the skin sample. Deformation and wave propagation are recorded with an ultrafast camera, enabling calculation of elastic modulus and viscoelastic properties based on deformation width and wave dynamics over micrometre scales and millisecond timescales, adapted from [165].

lose accuracy. In such cases, incorporating bending into inverse finite-element analysis improves accuracy [448]. Additionally, the method requires precise control over applied pressure and accurate measurement of deformations, which can be challenging in biological applications.

Another limitation arises in the characterisation of anisotropic biological tissues. While traditional bulge testing assumes isotropic properties, fibrous tissues exhibit directional stiffness variations, requiring more advanced analytical techniques such as Moiré interferometry or 2D image analysis [449–451]. Large-scale anisotropic features such as Langer's lines, which follow the natural orientation of collagen fibres in skin, can influence macroscopic mechanical response. However, for localised measurements on the micrometre scale, this directional variability is often averaged out unless specifically resolved by high-resolution mapping techniques.

Moreover, classical linear elastic models are insufficient to describe large deformations in complex materials, requiring the use of more sophisticated constitutive equations such as the neo-Hookean and Arruda-Boyce models [452,453].

6.1.2. BGT in soft and biological materials

Bulge testing has been extended to biological tissues, particularly in studying anisotropic properties of fibrous tissues. By analysing deformed profiles or using Moir interferometry, researchers have been able to determine strain components and membrane stresses along specific orientations [449–451]. These measurements yield quantitative values for directional elastic moduli and Poisson ratios, rather than just qualitative indications of stiffness.

A variation known as the hydrodynamic bulge test replaces direct deformation tracking with a simple flow-induced pressure measurement. In this method, a viscous liquid is pumped beneath the membrane, and the steady-state pressure drop for a given flow rate is recorded. From that pressure–flow relationship, one can calculate the tension at the membrane or the effective elastic modulus, eliminating the need for optical or contact deformation measurements [454]. This approach simplifies experimental setups and expands the applicability of bulge testing to a wider range of soft materials.

6.2. Flow elastography (FEL) main features

Flow elastography is an indentation technique that uses fluid (water or air) flow to deform the tissue. The fluid stream is delivered at a controlled pressure through a nozzle onto the material surface, causing a localised deformation [456]. Usually, the deformation of the material is tracked with optical or ultrasonic techniques [457,458] (see Section 3 and Fig. 12(b)).

By varying the flow rate and pressure, the magnitude of applied stress and the resulting strain rate can be varied, enabling characterisation across a range of mechanical regimes such as quasi-static indentation versus high-rate loading [459]. Additionally, the flow can be used intermittently to access the viscoelastic properties of the material, like the complex modulus (see Eq. (4)) [458,460]. In contrast to indentation with solid probes, the potential for sample damage, contamination, and influence of the probe material properties is lower [461–463]. The spatial resolution of these systems depends primarily on the cross-sectional size and shape of the indenting flow. The nozzle diameter typically ranges from 200 μm to 5 mm, which can be adjusted according to the desired spatial resolution.

6.2.1. FEL limitations

The precision and repeatability of the indentation depend on controlling flow conditions, nozzle geometry, and positioning stability. During jet impact, liquid can accumulate ("pool") at the contact site, potentially altering local mechanics. To minimise pooling, the jet is often delivered in short pulses or directed at a slight angle to encourage fluid runoff. In some setups, a synchronised suction or absorbent backing is placed beneath the sample to collect excess liquid immediately after impact. High-resolution measurements require careful calibration of both the jet parameters and any liquid-removal system.

6.2.2. FEL in soft and biological materials

Previous studies have focused on developing a non-contact ultrasonic indentation system, in which a water jet serves both as the indenter and as the coupling medium for high-frequency focused ultrasound. This system has been employed to determine Young's modulus and Poisson's ratio of phantoms and articular cartilage specimens [464–466]. A miniaturised water-jet ultrasound indentation probe was later designed to assess cartilage degeneration in bovine samples [467].

In other works [457] a 200 μm water jet with speeds ranging from 0.4–9.7 m/s has been used to indent polyacrylamide (PAA) and polydimethylsiloxane (PDMS) gels, using optical methods to track deformation. The lateral resolution achieved in these measurements was in the range of 50–500 μm. Furthermore, microjets generated by thermocavitation can reach speeds of approximately 100 m/s and can be used to deform the skin. The resulting deformation is captured with a high-speed camera, allowing for the estimation of the skin's elastic and loss moduli [165] (see Fig. 12(c)). These measured mechanical parameters inform the jet speed required to achieve a desired penetration depth and payload deposition for needle-free drug delivery.

Water flow elastography (WaFE) was proposed for quantifying Young's modulus in silicone samples and porcine organs for minimally invasive surgery [468]. This device uses suction to fix the probe's position and jets made of saline solution to induce indentation, and the results were validated using finite element simulations.

Flow elastography has found significant application in ophthalmology. In air-puff tonometry, a controlled burst of air deforms the cornea and the resulting displacement is used to infer intraocular pressure [469]. Building on this principle, an air-jet indentation system integrated with OCT, was developed to enable high-resolution, non-contact imaging of soft-tissue elasticity [470,471]. More recently, this OCT-coupled air-jet approach has been adapted for assessing the mechanical properties of diabetic wounds, offering a non-invasive tool for monitoring tissue healing [472,473].

Furthermore, a scanning air puff tonometer has been introduced to map viscoelastic properties of nearly flat soft materials [463] (See Fig. 12(b)). With air stream diameters down to 10 μm, the scanning air puff tonometer enables high-resolution, non-contact measurements. Further developments include integrating nanometric tips to create an "air jet AFM," using Nomarski mode tonometry to enhance deformation accuracy, and employing high-speed cameras to improve response curve resolution, especially for highly elastic materials.

A comparison of both solid-contact methods (Section 5) and liquid-contact methods (Section 6) in terms of spatial resolution, temporal resolution, and interrogation area is provided in Fig. 13.

7. Cavitation rheology (CR) main features

Cavitation is a process in which rapid pressure changes in liquids lead to the formation, growth, and collapse of bubbles within the liquid medium [474]. This occurs when the pressure drops below the vapour pressure, forming small vapour pockets that collapse after the pressure recovers. During the collapse, a shock wave is often produced and high-speed liquid jets are observed [475–477]. Such collapse and jet formation are known in engineering, for their damage potential in structures such as ship propellers, pipelines and biological structures like the brain [478–480]. In contrast, more positive uses of cavitation have been found to generate liquid jets for needle-free injections [10,481,482].

Cavitation has also been used to assess the mechanical properties of soft matter, a technique known as cavitation rheology [82] (see Fig. 14). CR uses the injection of pressurised fluid into gels using a needle (needle-induced cavitation or NIC) [483,484]. This concept has been expanded to include various methods beyond NIC, such as laser-induced cavitation (LIC), acoustic-induced cavitation (AIC), and shock-wave-induced

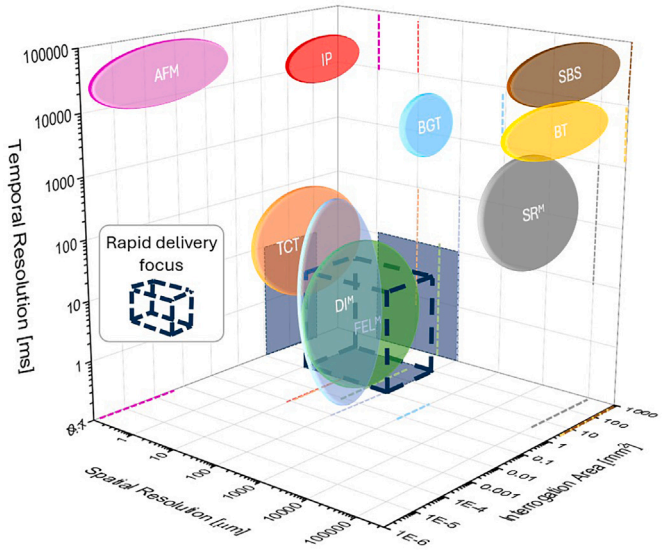


Fig. 13. Comparison of contacting solid and liquid methods based on spatial resolution, temporal resolution, and interrogation area. Each coloured ellipsoid denotes the typical operating envelope of a specific method: AFM and IP offer nanometre-to-micrometre spatial resolution over sub-micron contact areas but require slow, second-scale acquisitions; SR and BT probe millimetre-scale volumes with millisecond-to-second response times yet deliver only coarse spatial detail; BGT occupies intermediate spatiotemporal regimes; TCT can be actuated with piezoelectric materials providing 10–100 ms temporal resolution; FEL uses fluid jets to deform millimetre-scale areas in sub-millisecond pulses; and DI spans mid-range speeds and contact sizes. This visualisation highlights the trade-offs among probing modalities and guides the selection of appropriate methods for dynamic, multiscale mechanical characterisation of soft materials. SR, IA and TR ranges correspond to typical literature demonstrations for each technique, unless otherwise indicated. Entries for which the reported range is taken primarily from manufacturer specifications are labelled "M".

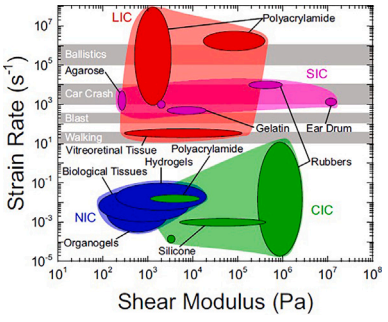


Fig. 14. Cavitation rheology techniques and their capabilities mapped in shear modulus–strain rate space. The coloured envelopes show the accessible ranges of four cavitation methods: needle-induced cavitation (NIC, blue), capillary-induced cavitation (CIC, ForestGreen), laser-induced cavitation (LIC, red), and shock-induced cavitation (SIC, magenta). Grey bands mark typical strain rates for common events: walking, blasts, ballistic impacts and car crashes. Data points and sub-regions indicate representative materials—from soft organogels and hydrogels (agarose, polyacrylamide, gelatin) and biological tissues (vitreoretinal tissue, eardrum) to stiffer rubbers and silicones—highlighting how cavitation rheology bridges micro- to macroscopic mechanical characterisation. Reprinted from reference [82], which is licensed under CC BY-NC-ND 4.0.

cavitation (SIC) [82]. These techniques exploit the fast dynamics and small scales in which cavitation occurs to assess material properties, particularly valuable for soft materials and biological tissues, providing assessments at higher strain rates (ranging from 10^3 to 10^8 s^{-1}) compared to contacting solid methods [83].

7.1. CR limitations

NIC requires slow pressure increases to maintain deformation assumptions, leading to strain rates on the order of 10^{-1} s^{-1} [83,483]. Additionally, while LIC enables assessment of material properties at higher shear rates, it may be too invasive for skin assessment, potentially requiring optical fibre insertion and posing risks of tissue damage due to heating and cavitation jet generation afterward. Similarly, AIC (acoustic-induced cavitation) uses high-frequency ultrasound—typically in the range of hundreds of kilohertz to several megahertz—to nucleate and drive cavitation bubbles. These high frequencies (10^5–10^7 s^{-1}) allow rapid oscillatory loading of tissue, particularly in *ex vivo* settings. However, because cavitation can produce strong local stresses and microjets, the risk of tissue damage (e.g., micro-tears) must be evaluated for each application and tissue type [485,486]. In SIC, while strain measurements can be conducted using a combination of Digital Image Correlation (DIC) see Section 3.2.1 and high-speed imaging, its application is primarily limited to *ex vivo* experiments due to the potential tissue damage caused by shock waves [487].

7.2. CR in soft and biological materials

NIC has been used to measure the mechanical properties of *ex vivo* rat skin using the critical yielding pressure, which correlates with the elastic modulus of the skin [486,488]. NIC has also been employed to evaluate the mechanical properties of bovine vitreous *in vivo* and *ex vivo*, revealing that the Young's modulus is lower when measured *ex vivo* [489]. NIC has also been instrumental in assessing the elastic modulus and anisotropy of the eye lens across different length scales, from single cells to millimetres [490].

LIC has been successfully used to analyse agar skin phantoms by tracking elastic wave characteristics following laser-induced cavitation [83]. While LIC can probe eye tissue non-invasively—since the cornea is transparent and the laser pulse can be focused inside without damaging the surface—applying the same approach to skin requires ablating or otherwise disrupting the epidermal surface to generate a cavitation bubble, making it invasive.

AIC has been applied to agarose gels of varying concentrations by monitoring bubble dynamics and applying a Kelvin-Voigt model,

$$\sigma(t) = E\epsilon(t) + \eta\frac{d\epsilon(t)}{dt}, \tag{5}$$

where σ is stress, ϵ is strain, E is the Young's modulus, and η is the viscous damping coefficient. This fitting procedure allows extraction of both elastic and viscous parameters to characterise viscoelastic behaviour [486]. Additionally, mechanical properties of *ex vivo* animal eye globes have been measured by inducing microbubbles at different positions and exposing them to acoustic radiation, allowing Young's modulus determination [491]. Cavitation bubbles sonophoresis—which is the use of ultrasound to enhance transdermal drug delivery by creating temporary pores in the skin—has been shown to correlate with mechanical properties of skin, as a new approach to simultaneously evaluate skin mechanics and improve drug delivery [492,493].

SIC has been employed to measure strain up to 0.5 within an area of 0.18 mm^2 within 24 microseconds following bubble collapse [487]. This technique is particularly valuable for measuring mechanical properties at strains up to 10^4 s^{-1} and has been used in *ex vivo* studies to evaluate potential tissue damage from shock waves in ophthalmic procedures like posterior capsulotomy and vitreolysis [494].

8. Experimental variability and sample handling considerations

This section summarises the main experimental aspects of relevance for the accurate mechanical characterisation of soft tissues. Besides intrinsic sample differences, each measurement technique can be influenced by external factors, related to sample handling before and during measurements [495]. The most salient are:

- **Storage conditions**, including the medium in which the tissue is kept, light exposure, temperature, and humidity, can significantly affect the mechanical properties of biological samples [496–498]. Variations in these parameters may lead to uncontrolled changes in tissue hydration and composition, which in turn influence properties such as stiffness and elasticity. It is therefore critical to standardise storage and measurement protocols—such as those outlined by ASTM F2150 ("Standard Guide for characterisation and Testing of Biomaterials") to minimise these effects.
- **Sample Hydration.** The degree of hydration determines and causes variation in tissue mechanics [499,500]. Changes in water content can alter the viscoelastic properties of soft tissues, resulting in discrepancies between measurements [501]. Ensuring consistent hydration levels throughout the experimental process is essential for obtaining reliable data.
- **Buffers and other chemical agents** used during sample preparation can modify the structure and mechanical behaviour of tissues [499,502]. The choice of buffer, its pH, and ionic composition can impact the observed properties [503,504]. Detailed documentation of these conditions is necessary to interpret and compare the results accurately.
- **Mounting and Mechanical Preconditioning.** The choice made to mount or fix samples can affect their response under testing, e.g., if they are stretched, kept loose, or preconditioned mechanically [505]. Variations in mounting can introduce stress anisotropies or pre-strain in the tissue, thereby influencing the measurement of parameters such as the Young's modulus and shear modulus [506].
- **Time between Sample Collection to Measurement.** The interval between sample collection and measurement, including any *ex vivo* storage time, is another critical factor [507]. As tissues undergo biochemical and structural changes over time, delays in measurement can result in significant differences in mechanical properties [498]. Therefore, reducing and reporting the time elapsed from sample harvesting to testing is crucial.
- **Illumination Exposure.** In optical and fluorescence-based measurements, the total duration and intensity of lamp or laser illumination can affect tissue properties and signal quality. Prolonged exposure may induce photothermal heating, leading to local changes in hydration and mechanical stiffness, or photobleaching of fluorophores, reducing contrast and accuracy. To minimise these artefacts, it is essential to record cumulative illumination time and power density, use pulsed or attenuated light where possible, and validate that exposure levels do not alter the sample's mechanical response.

9. Concluding remarks and outlook

We have provided a non-exhaustive review of the state-of-the-art experimental techniques for measuring the mechanical properties of soft materials. Our focus on the skin and eye is justified by their suitability for topical drug delivery, particularly due to the relatively thin and accessible nature of their barrier membranes. We have discussed a diverse range of methods—including optical, acoustic, and hybrid approaches—that span multiple spatial and temporal scales. Each technique presents its own strengths and limitations, from the high-resolution capabilities of speckle interferometry and digital holography to the non-invasive, *in vivo* potential of ultrasound elastography. By comparing methodologies that probe both macroscopic and microscopic deformations, we have highlighted the critical role that measurement parameters and experimental conditions play in accurately characterising tissue mechanics. One plausible approach is combining two or more techniques that can provide complementary information.

The integration of advanced imaging modalities with innovative sensing approaches holds great promise for capturing real-time, multiscale mechanical responses in biologically relevant tissues. By leveraging developments in portable instrumentation, high-throughput data acquisition, and automated analysis pipelines, future systems will be capable of mapping tissue stiffness and viscoelasticity *in situ* with minimal invasiveness. Additionally, ongoing progress in miniaturisation and ultrahigh-speed imaging technologies will further enhance our ability to resolve microscale heterogeneity and anisotropic mechanical behaviour, bringing comprehensive biomechanical profiling closer to routine clinical and research applications.

To give a visual summary, Fig. 15 presents a schematic overview of all techniques discussed in this review. The methods are organised by whether they provide mechanical properties directly or indirectly, and by whether they are applicable *in vivo*, *ex vivo* only, or potentially adaptable to *in vivo* use. In the context of this figure, methods placed in the *in vivo* rows are those that have already been implemented in *in vivo* experiments on the eye, skin or other soft tissues, even if their current use is largely confined to research settings rather than routine clinical practice. Likewise, the direct/indirect labels follow the definitions introduced earlier in the review: a) direct methods can provide output mechanical properties (for example optical or acoustic elastography, indentation or shear rheology); b) indirect methods act primarily as readouts of geometry, motion or contrast and must be combined with a separate loading modality or mechanical model to obtain quantitative material parameters (for example, optical or acoustic imaging when used as the readout module in elastography). This map highlights how optical, acoustic, solid-contact, fluid-contact and cavitation-based methods occupy distinct but complementary roles, helping researchers quickly identify which families of techniques are most appropriate for a given application. The map can also be used in combination with the radar plot presented in Fig. 16.

Reducing experimental variability remains an urgent challenge. Rigorous standardisation of experimental conditions and thorough documentation of sample handling protocols are essential to minimise variability and improve the reliability of mechanical measurements in soft tissues. Future studies should aim to systematically investigate the combined influence of factors such as storage conditions, hydration, buffer composition, sample mounting, and timing between collection and measurement, and to develop robust guidelines that ensure consistency across different experimental setups. Additionally, optimising calibration protocols will strengthen the reliability of *in situ* measurements.

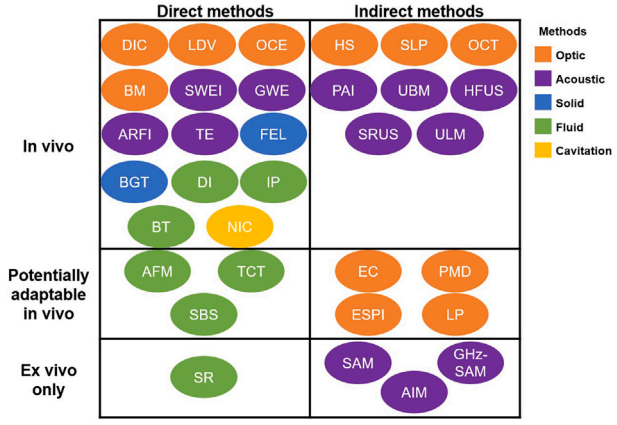


Fig. 15. Schematic classification of the techniques discussed in this review. Methods are arranged by whether they directly provide mechanical properties (left column) or act as indirect imaging/readout modalities that must be coupled to a separate loading or modelling approach (right column), and by their reported implementation *in vivo*, their potential adaptability to *in vivo* use, or current restriction to *ex vivo*-only experiments (rows). Colours indicate the underlying physical principle: orange – optical; purple – acoustic; blue – solid-contact; green – fluid-contact; yellow – cavitation-based. Abbreviations are defined in the main text.

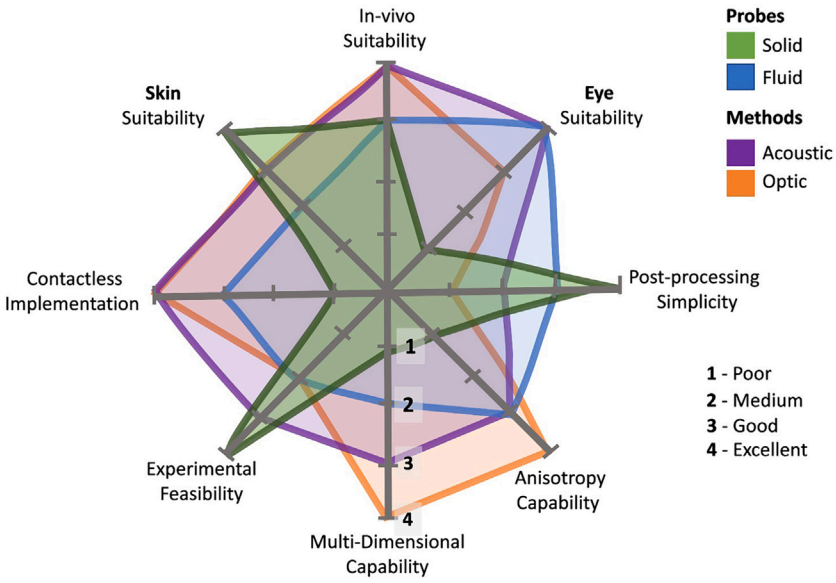


Fig. 16. Radar plot showing the main capabilities of each type of measurement technique. The techniques are grouped in four categories: contacting solid methods, contacting fluid methods, acoustic methods and optical methods.

Finally, we note that the combination of complementary measurement approaches offers a powerful multi-scale, multi-timescale characterisation of soft tissues. High-speed imaging and instrumented indentation/impact tests capture the transient viscoelastic response of tissue to rapid loadings (0.1–100 ms), while non-contact elastography (e.g., optical coherence or ultrasound-based methods) and surface deformation techniques provide spatial maps of stiffness and hysteresis at the millimetre scale. Combining these data with local permeability or retention assays and with computational constitutive models allows direct correlation between mechanical stimuli, local microstructure, and transport processes. This multi-scale, multi-physics synergy will improve mechanistic understanding and permit rational tuning of device parameters (e.g., microneedle geometry, jet velocity) and formulation rheology to enhance uptake and reduce tissue damage. Ultimately, such an integrated approach should speed the translation of biomechanical insights into safer and more effective ocular and transdermal drug-delivery strategies.

Besides the traditional role of universities and research institutes, we believe that established and startup companies have the potential to accelerate the development and adoption of advanced measurement techniques. In sum, the continued development and validation of these methodologies will be instrumental in advancing scientific research, improving diagnostic accuracy, and providing personalised therapeutic interventions in biomedical applications.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT to improve the clarity, grammar, and style of some parts of the text. After using this tool/service, the authors reviewed and edited the content, and assume full responsibility for the content of the publication.

Declaration of competing interest

The authors declare the following financial interests/personal relationships that may be considered as potential competing interests:

David Fernandez Rivas reports that financial support was provided by the NWO Talent Programme Vidi TTW. David Fernandez Rivas and Miguel Quetzeri report that financial support was provided by the European Research Council (ERC). Miguel Quetzeri reports that financial support was provided by DGAPA bprograma de Incorporacion de Jovenes Academicos de Carrera (SIJA) and through the grant PAPIIT-UNAM IA101025. David Fernandez Rivas reports a relationship with FlowBeams that includes: board membership, employment, and equity or stocks. David Fernandez Rivas and Miguel Quetzeri have patent #Fernandez Rivas, D. & Quetzeri-Santiago, M. A., 2023. Material method characterisation. NL2030901B1;WO2023153926A1. pending to the University of Twente. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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